

Ontological Constraints and the Mathematical Foundations of the Future Information Field in Future Convergence Theory

A Unified Reconstruction of Possibility, Quantum Allowance, Gravitational Fixation,
and Projection

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June 29, 2026

Abstract

This paper reconstructs Future Convergence Theory (FCT), originally established through the unified axiomatic framework presented in Paper 93, as a more fundamental theory of ontological constraints.

Paper 93 introduced the fundamental hierarchical sequence

$$M \rightarrow F \rightarrow Q \rightarrow X,$$

through which the Meaning Generation Template, the Future Information Field, the Quantum Allowance Layer, and the three-dimensional Manifestation Layer were defined within a unified framework. However, several foundational questions remained unresolved: how the Future Information Field acquires the capacity for manifestation, how possibility itself originates, and why quantum phenomena, gravity, the four fundamental interactions, dark matter, and dark energy possess ontological necessity.

In this paper, possibility is not treated as a pre-existing set of candidate states. Instead, possibility is defined as the reachable domain that remains after ontological constraints

$$C_{\Omega}$$

act upon the undifferentiated preservation structure

$$\Omega,$$

eliminating regions that are ontologically impossible. Consequently, the Future Information Field is not regarded as a completed possibility space existing a priori, but rather as a preserved reachable structure generated through ontological constraints.

Furthermore, the Quantum Allowance Layer is redefined as an ontological margin of indeterminacy that prevents complete fixation, whereas gravity is reformulated as the ontological fixation operator that prevents complete independence and anchors manifested states within three-dimensional spacetime. Likewise, the strong interaction, electromagnetic interaction, and weak interaction are reinterpreted as distinct classes of ontological constraints that respectively prohibit complete separation, complete isolation, and complete invariance.

The paper further integrates the Future Information Tensor, the Constraint Tensor, the Meaning Density Tensor, and the Premise Update Tensor into a single preservation tensor theory, thereby describing cognition, meaning, language, and the physical universe as distinct projections of one common preservation structure. Under this framework, cognitive theory and cosmology are no longer independent disciplines but become different manifestations of a single geometry of ontological constraint interactions.

Finally, dark matter is formulated as a local non-material projection of ontological constraints, dark energy as the cosmological mean constraint, black-hole information preservation as non-vanishing constraint preservation on the Future Information Tensor, and

gravitational-wave residuals as observational windows into the re-equilibration process of ontological constraints. Through these developments, the unresolved foundations remaining after Paper 93 are unified under the single principle of ontological constraint, reconstructing Future Convergence Theory as a mathematical ontological unified theory equipped with axioms, theorems, principles of action, and empirical falsifiability.

Keywords: Future Convergence Theory, Future Information Field, Ontological Constraint, Undifferentiated Preservation Structure, Quantum Allowance Layer, Ontological Fixation, Future Information Tensor, Meaning Density Tensor, Projection, Dark Matter, Dark Energy, Mathematical Ontology, Fundamental Physics

1 Introduction: Issues Remaining After Paper 93

1.1 Purpose of This Paper

The purpose of this paper is to reconstruct Future Convergence Theory (FCT), originally established through the unified axiomatic framework presented in Paper 93, as a deeper theory of ontological constraints.

In Paper 93, FCT was defined by the following fundamental hierarchical sequence:

$$M \longrightarrow F \longrightarrow Q \longrightarrow X$$

where

M

denotes the Meaning Generation Template,

F

denotes the Future Information Field,

Q

denotes the Quantum Allowance Layer,
and

X

denotes the three-dimensional Manifestation Layer.

Within this framework, Paper 93 provided a unified description of gravity, quantum theory, black holes, dark matter, dark energy, the early universe, gravitational waves, the Page curve, the Casimir effect, the cosmic microwave background (CMB), and observational criteria for empirical falsification.

Nevertheless, several fundamental issues remained unresolved.

First, although the Future Information Field was defined as that which preserves information, its capacity for manifestation was not fully axiomatized.

Second, although the Quantum Allowance Layer was introduced as the layer that filters manifestable states, the origin of possibility itself remained undefined.

Third, although gravity was redefined as the constraint governing manifestation, the ontological necessity of the other three fundamental interactions—the strong interaction, the weak interaction, and the electromagnetic interaction—was not derived within the framework of FCT.

Fourth, although dark matter and dark energy were interpreted as constraints originating from the Future Information Field, the conditions distinguishing their manifestation as material

entities from their appearance as non-material geometrical constraints remained insufficiently specified.

Fifth, although FCT provides a comprehensive cosmological framework, unresolved problems in particle physics—including baryon asymmetry, CP violation, the Standard Model gauge groups, neutrino masses, and the hierarchy problem—have not yet been treated as central theoretical objects.

This paper reconstructs all of these unresolved issues through a single foundational concept:

Ontological Constraint.

1.2 Central Thesis of This Paper

The central thesis of this paper is the following:

Possibility does not originally exist within the Future Information Field.

More precisely,

Possibility is the reachable domain that remains after an undifferentiated preservation structure is acted upon by ontological constraints.

Accordingly, this paper rejects the conventional sequence

Possibility $\longrightarrow$ Selection $\longrightarrow$ Manifestation.

Instead, this paper adopts the following ontological sequence:

Undifferentiated Preservation Structure
 $\longrightarrow$ Ontological Constraint
 $\longrightarrow$ Elimination of Impossible Regions
 $\longrightarrow$ Reachable Domain
 $\longrightarrow$ Quantum Allowance
 $\longrightarrow$ Manifestation

Within this framework, possibility is not an initial premise.

Rather, possibility is defined only as the residual reachable domain remaining after ontological constraints eliminate regions that cannot exist.

Consequently, possibility is not the most fundamental element of Future Convergence Theory.

Neither information nor purpose occupies the deepest ontological level.

Instead, this paper introduces the symbol

Ω

to denote the most fundamental entity, referred to as the *Undifferentiated Preservation Structure*.

The Undifferentiated Preservation Structure is a preservational substrate that has not yet differentiated into possibility, matter, spacetime, quantum states, or meaning.

The Future Information Field emerges only after this undifferentiated preservation structure is subjected to ontological constraints.

That is,

$$\Omega \xrightarrow{C_\Omega} F.$$

Here,

$$C_\Omega$$

denotes the set of ontological constraints.

1.3 What Are Ontological Constraints?

Within this paper, a constraint is not merely a restriction.

Nor is it an external condition imposed upon an object.

Instead, an ontological constraint is defined as

a configuration of relations that renders alternative modes of existence impossible.

Accordingly, constraints do not perform selection.

Rather, they determine what cannot exist.

This definition completely reverses the conventional interpretation of possibility.

Traditionally,

$$\text{Possibility} \longrightarrow \text{Selection}$$

is assumed.

In contrast, this paper argues

The elimination of impossible regions gives rise to possibility.

Possibility is therefore not freedom preceding constraints.

It is the residual structure remaining after constraints have acted.

Under this viewpoint, cosmic manifestation is not the selection of one possibility from infinitely many alternatives.

Instead, manifestation is the process whereby ontological constraints act upon the Undifferentiated Preservation Structure, eliminate impossible regions, preserve the reachable domain, pass it through the Quantum Allowance Layer, and finally project it into the three-dimensional Manifestation Layer.

$$\Omega \xrightarrow{C_\Omega} F \xrightarrow{Q} \mathcal{A} \xrightarrow{P_X} X$$

where

$$\mathcal{A}$$

denotes the reachable domain.

1.4 Relationship to Paper 93

In Paper 93, the fundamental hierarchy of Future Convergence Theory was defined as

$$M \longrightarrow F \longrightarrow Q \longrightarrow X.$$

The present paper does not reject this hierarchy.

Rather, it preserves the original sequence while introducing an ontological layer preceding it.

The revised hierarchy becomes

$$\Omega \longrightarrow C_\Omega \longrightarrow F \longrightarrow Q \longrightarrow X.$$

This additional ontological layer resolves one of the principal unresolved questions left open in Paper 93:

Why does the Future Information Field possess the capacity for manifestation?

The answer proposed in this paper is that the Future Information Field does not inherently contain possibility.

Instead, the Future Information Field emerges only after the Undifferentiated Preservation Structure has been acted upon by ontological constraints, thereby becoming a preservational structure capable of manifestation.

Accordingly,

$$F$$

is not a completed possibility space.

Rather,

$$F$$

is a preservation structure from which reachable domains may be generated through ontological constraints.

For this reason, the present work should not be regarded as a revision that supersedes Paper 93.

Nor does it invalidate the theoretical foundations established there.

Instead, this paper supplements the foundation of Paper 93 by introducing the ontological constraint layer, thereby providing a more rigorous definition of both the origin and the authority of the Future Information Field.

1.5 Redefinition of the Quantum Allowance Layer

In Paper 93, the Quantum Allowance Layer was introduced as an intermediate layer that selects which states within the Future Information Field may proceed toward manifestation in three-dimensional reality.

The present paper refines this definition.

The Quantum Allowance Layer is not a layer that presupposes possibility.

Rather, it preserves the reachable domain remaining after ontological constraints have eliminated impossible regions, while preventing complete fixation.

Thus, quantum allowance is defined as

the ontological constraint that prohibits complete fixation.

If every state were completely fixed, neither creation, change, selection, nor superposition could exist.

Conversely, if nothing were fixed, neither structure, matter, spacetime, nor observation could emerge.

The Quantum Allowance Layer therefore occupies the ontological region between complete fixation and complete freedom.

This condition is expressed as

$$0 < W < \infty,$$

where

$$W$$

denotes the Quantum Creative Margin.

This margin is not an undefined ambiguity.

Rather, it is the indispensable ontological region of non-fixation required for manifestation itself.

1.6 Redefinition of Gravity

Paper 93 redefined gravity not as an interaction between particles but as the manifestation constraint that preserves the mapping from the Future Information Field into the three-dimensional Manifestation Layer.

The present paper retains this interpretation while refining it further.

Gravity is redefined as

the ontological constraint that prohibits complete independence.

If every manifested state were completely independent, they could not constitute a single universe.

Neither spacetime, geometry, causality, nor a common ontological arena could exist.

Consequently, for the three-dimensional Manifestation Layer to emerge as one coherent universe, a constraint must exist that forbids complete independence among manifested states.

That constraint is gravity.

Gravity is therefore not a secondary force generated by matter.

Rather, it is the global ontological fixation through which manifested states are bound into a common spacetime.

Its manifestation in spacetime is expressed by

$$G_{\mu\nu} \sim \Pi(C_{\mu\nu}),$$

where

$$C_{\mu\nu}$$

is the Manifestation Constraint Tensor, and

$$\Pi$$

denotes the projection operator into three-dimensional spacetime.

1.7 Repositioning the Four Fundamental Interactions

This paper does not regard the four fundamental interactions as primitive forces existing from the outset.

Instead, they are interpreted as ontological constraint types that eliminate forms of existence incompatible with manifestation.

Specifically,

Quantum	:	prohibits complete fixation,
Gravity	:	prohibits complete independence,
Strong Interaction	:	prohibits complete separation,
Electromagnetic Interaction	:	prohibits complete isolation,
Weak Interaction	:	prohibits complete invariance.

Under this formulation, physical laws are no longer interpreted as forces acting within the universe.

Instead, they become ontological constraints that eliminate modes of existence incompatible with the emergence of the universe itself.

The strong interaction prevents local structures from becoming completely separated.

The electromagnetic interaction prevents manifested structures from becoming entirely unrelated.

The weak interaction prevents states from remaining perfectly invariant.

Gravity prevents manifested structures from existing in complete independence.

Quantum allowance prevents absolute fixation.

Only through these five ontological constraints can three-dimensional manifestation simultaneously possess stability, change, interaction, locality, global geometry, and a finite margin of indeterminacy.

2 Abandoning Possibility as a Primitive Assumption — Deriving the Reachable Domain through Ontological Constraints

2.1 Problem Statement

Not only in modern physics, but also in many branches of ontology, cosmology, and information theory, *possibility* is implicitly assumed as a primitive concept.

A common conceptual structure is

$$\text{Possible States} \longrightarrow \text{Selection} \longrightarrow \text{Reality}.$$

In quantum mechanics, superposed states are regarded as possibilities.

In statistical mechanics, the state space is given as the set of possible states.

In cosmology, possible cosmic histories are discussed based on given initial conditions.

Despite their differences, these theories share a common unresolved question:

Where does possibility itself originate?

This paper does not assume possibility as something that exists a priori.

Instead, this chapter demonstrates that possibility can be defined only as the consequence of ontological constraints.

By doing so, the most fundamental axioms of Future Convergence Theory are reconstructed.

2.2 The Problem with Taking Possibility as an Axiom

Suppose possibility is assumed from the outset.

Then one begins with a set of possible states,

$$P = \{p_1, p_2, \dots, p_n\}.$$

However, this assumption immediately raises a deeper question.

Who defines this set?

By what principle is it determined?

Why do these possibilities exist instead of others?

Thus,

$$P$$

is not itself explained.

Rather, it becomes the unexplained starting point of explanation.

From the standpoint of ontology, this constitutes a termination of explanation.

Ontology should explain the origin of possibility rather than presupposing it.

Accordingly,

$$P$$

must be removed from the primitive axioms.

2.3 The Undifferentiated Preservation Structure

We therefore define the most fundamental ontological object as

$$\Omega.$$

Definition

$$\Omega = \text{Undifferentiated Preservation Structure.}$$

The Undifferentiated Preservation Structure is the preservational substrate that is

- neither matter,
- nor energy,
- nor spacetime,
- nor a quantum state,
- nor possibility,
- nor meaning.

The crucial point is that

$$\Omega$$

contains no notion of "possible states."

That is,

$$\Omega \neq P.$$

The Undifferentiated Preservation Structure contains no distinctions.
Consequently, the very concept

“this state is possible”

has no meaning at this stage.

2.4 Ontological Constraints

We now introduce the concept of ontological constraints.

Definition An ontological constraint is an operator

$$C_{\Omega} : \Omega \rightarrow \Omega'$$

whose function is to determine regions that are ontologically impossible. Conventionally, constraints are regarded as restrictions imposed upon freedom. In this paper, however, an ontological constraint is defined as

an operation that fixes regions incapable of existence.

Consequently,
 an ontological constraint does not determine
 what is selected,
 but rather
 what cannot exist.
 It is therefore not a selection operator,
 but an elimination operator.

2.5 Impossible Regions

After the action of ontological constraints,
 the preservation structure separates into two components:

$$\Omega = \Omega_{\text{reachable}} \cup \Omega_{\text{forbidden}}.$$

Here,

$$\Omega_{\text{forbidden}}$$

denotes the region declared ontologically impossible by the constraints.
 Conversely,

$$\Omega_{\text{reachable}}$$

is the residual preservation structure that has not been eliminated.
 This paper defines only this residual component as the
reachable domain.

2.6 Redefining Possibility

Traditionally,
 possibility has been understood as a pre-existing collection of candidate states.
 In contrast,
 this paper defines possibility as

$$P \equiv \Omega_{\text{reachable}}.$$

Thus,
 possibility is nothing more than the reachable domain remaining after impossible regions
 have been removed from the Undifferentiated Preservation Structure.
 Possibility is therefore not the cause.
 It is the result.
 This fundamentally changes the ontological status of possibility.

2.7 The Emergence of the Future Information Field

In Paper 93,

the Future Information Field was defined as the informational field enabling Future Projection.

The present paper redefines it as

$$F = \mathcal{F}(\Omega_{\text{reachable}}).$$

Here,

$$\mathcal{F}$$

is the structural mapping that transforms the reachable domain into the Future Information Field.

The Future Information Field is therefore not a field that stores possibilities.

Rather,

it is the informational preservation state of the reachable domain.

Consequently,

before the emergence of the Future Information Field,

the concept of possibility does not yet exist.

2.8 The Necessity of Projection

Projection is not an act of intention.

Nor is it a purpose.

Projection follows necessarily from the structural properties of the reachable domain.

Thus,

$$F \rightarrow Q \rightarrow X$$

does not occur because the Future Information Field "wishes" to manifest.

Instead,

once the Future Information Field exists as a preserved reachable structure,

Projection becomes its necessary structural consequence.

Projection is therefore not an act of will,

but a condition required for structural preservation.

2.9 Ontological Significance

The ontological hierarchy established in this paper is therefore

$$\Omega \longrightarrow C_{\Omega} \longrightarrow \Omega_{\text{reachable}} \longrightarrow F \longrightarrow Q \longrightarrow X.$$

Within this sequence,

possibility is absent at the beginning.

Possibility appears only after ontological constraints eliminate impossible regions.

The universe therefore does not originate from possibility.

Rather,

the universe is that which generates possibility through ontological constraints.

2.10 Conclusion of This Chapter

This chapter abandons ontologies that take possibility as their primitive assumption.

Instead,

it introduces

the Undifferentiated Preservation Structure,

$$\Omega,$$

together with

the ontological constraint,

$$C_{\Omega}.$$

As a result,

possibility is redefined as the reachable domain remaining after ontological constraints have acted.

Consequently,

the Future Information Field is not a field containing possibilities.

Rather,

it is the informational preservation state of the reachable domain generated through ontological constraints.

The next chapter axiomatizes the ontological constraint set

$$C_{\Omega},$$

and provides a mathematical definition of the conditions under which the Future Information Field comes into existence.

3 Axiomatization of the Ontological Constraint Set — Conditions for the Emergence of the Future Information Field

3.1 Purpose of This Chapter

In the previous chapter, possibility was removed from the foundations of ontology.

The present chapter introduces the ontological constraint set

$$C_{\Omega},$$

which acts upon the Undifferentiated Preservation Structure

$$\Omega,$$

as a mathematical object.

The objective of this chapter is to derive the Future Information Field not as an information field that exists a priori, but as a preservation structure that necessarily emerges as the result of ontological constraints.

3.2 The Ontological Constraint Set

Definition The ontological constraint set is defined as

$$C_{\Omega} = \{C_1, C_2, \dots, C_n\},$$

where each

$$C_i$$

is an operator acting upon the Undifferentiated Preservation Structure

$$\Omega$$

to generate regions of ontological impossibility.
Accordingly,

$$C_i : \Omega \rightarrow \Omega.$$

The crucial point is that

$$C_i$$

does not add information,
does not generate particles,
and does not create space.
Its sole operation is the fixation of impossible regions.

3.3 Fundamental Axioms of Ontological Constraints

The ontological constraint set satisfies the following five axioms.

Axiom 1 (Non-Generative Axiom) Constraints generate nothing.

$$C_i(\Omega) \subseteq \Omega.$$

Ontological constraints never enlarge the preservation structure.
They eliminate rather than create.

Axiom 2 (Non-Selection Axiom) Constraints do not select states.

They determine only regions that cannot exist.
Accordingly,

$$C_i \neq \text{Selection}.$$

Axiom 3 (Preservation Axiom) After the action of constraints, the preservation structure itself is never destroyed.

Instead,

$$\Omega = \Omega_{\text{reachable}} \cup \Omega_{\text{forbidden}}$$

always holds.

Constraints separate modes of existence rather than annihilating existence itself.

Axiom 4 (No Requirement of Reversibility) Ontological constraints are not required to be reversible.

That is,

$$C_i^{-1}$$

need not exist.

This follows from the fact that manifestation itself generally contains irreversible processes.

Axiom 5 (Consistency Axiom) Different ontological constraints need not commute.

Thus,

$$C_i \circ C_j = C_j \circ C_i$$

is not required.

However, the final reachable domain must never become empty.

Therefore,

$$\Omega_{\text{reachable}} \neq \emptyset$$

must always hold.

3.4 The Action of Ontological Constraints

The total action of the constraint set is represented by the composite operator

$$\mathcal{C} = C_n \circ \dots \circ C_2 \circ C_1.$$

Then

$$\Omega' = \mathcal{C}(\Omega)$$

defines the preservation structure after constraint application.

At this stage,

$$\Omega'$$

is still not the Future Information Field.

The concept of information has not yet emerged.

Only an ontologically admissible preservation structure exists.

3.5 Impossible Regions

The action of the constraint operator separates the preservation structure into two components:

$$\Omega' = (\Omega_{\text{reachable}}, \Omega_{\text{forbidden}}).$$

The forbidden region should not be interpreted as non-existence.

Rather, it denotes the region excluded from becoming an object of manifestation by ontological constraints.

Thus,

“ontologically impossible”

does not mean

“non-existent.”

It means

“incapable of manifestation.”

This distinction is fundamental for the ontology developed in this paper.

3.6 The Emergence of the Future Information Field

The Future Information Field is not identical with the reachable domain itself.

It is defined as

$$F = \Phi(\Omega_{\text{reachable}}).$$

Here,

$$\Phi$$

denotes the preservation mapping that transforms the reachable preservation structure into an informational field.

The Future Information Field is therefore not a possibility space.

Rather,

it is the informational preservation state of the preservation structure remaining after ontological constraints have acted.

Accordingly,

the Future Information Field cannot be defined independently of ontological constraints.

3.7 What Is Information?

The concept of information is redefined in this paper.

Information is neither data,

nor meaning.

Information is that structure which remains preserved after ontological constraints have acted.

Thus,

$$\text{Information} \neq \text{Structure} - \text{Destroyed Structure},$$

but rather,

$$\text{Information} = \text{Preserved Reachable Structure}.$$

Under this definition,

the Future Information Field becomes the informational representation of the preserved structure.

3.8 Constraint Density

Ontological constraints possess a density.

Define the constraint density by

$$\rho_C = \frac{|\Omega_{\text{forbidden}}|}{|\Omega|}.$$

Then

$$0 \leq \rho_C < 1.$$

If

$$\rho_C = 0,$$

no ontological constraints exist,

and therefore no reachable domain can be defined.

Conversely,
if

$$\rho_C = 1,$$

no reachable domain remains,
and consequently the Future Information Field cannot emerge.
Hence,
the necessary condition for the existence of the Future Information Field is

$$0 < \rho_C < 1.$$

This implies that neither complete freedom nor complete constraint can generate an informational field.

3.9 The Ontological Equilibrium Condition

The necessary condition for the existence of the Future Information Field is the coexistence of both reachable and forbidden regions.

Accordingly,
the following conditions define the *Ontological Equilibrium Condition*:

$$\Omega_{\text{reachable}} \neq \emptyset,$$

and simultaneously,

$$\Omega_{\text{forbidden}} \neq \emptyset.$$

Only under this equilibrium can the Future Information Field exist as a preserved informational structure.

3.10 Connection to the Quantum Allowance Layer

Ontological constraints alone do not produce manifestation.

The Future Information Field remains a preservation structure.

Manifestation requires an additional layer capable of preserving the reachable domain in an unfixed state.

This layer is the Quantum Allowance Layer.

The ontological sequence therefore becomes

$$\Omega \longrightarrow C_\Omega \longrightarrow F \longrightarrow Q.$$

Quantum allowance is not the origin of possibility.

Rather,

it preserves the reachable domain generated by ontological constraints without allowing complete fixation, thereby preparing the structure for manifestation.

3.11 Conclusion of This Chapter

This chapter has axiomatized the ontological constraint set

$$C_\Omega$$

as the necessary condition for the emergence of the Future Information Field.

Consequently,

the Future Information Field is no longer interpreted as an information field existing from the outset.

Instead,

it is defined as the informational preservation state of the reachable structure remaining after the Undifferentiated Preservation Structure has been subjected to ontological constraints.

Furthermore,

information itself has been redefined as preserved reachable structure.

The next chapter further develops the Future Information Field as an ontological preservation structure and unifies it mathematically with the Future Information Tensor, the Meaning Density Tensor, and Premise Update Theory.

4 Redefinition of the Future Information Field — Ontological Preservation Structures and the Future Information Tensor

4.1 Purpose of This Chapter

In the previous chapter, we demonstrated that the Future Information Field emerges as the result of ontological constraints

$$C_{\Omega}$$

acting upon the Undifferentiated Preservation Structure

$$\Omega.$$

Nevertheless, the Future Information Field is still described as an "information field."

A fundamental question therefore remains unresolved:

What is information?

In this paper, information is defined neither as data, meaning, nor observational outcomes.

Rather, the Future Information Field is not merely preserved information; it is the ontological structure that performs preservation itself.

Accordingly, this chapter redefines the Future Information Field as an ontological preservation structure and introduces the Future Information Tensor.

Furthermore, the Meaning Density Tensor, the Constraint Tensor, and Premise Update Theory are mathematically unified within a single preservation tensor framework.

4.2 Reconsidering the Concept of Information

In conventional information theory, information is generally defined as either a state or a difference between states.

For example, Shannon information is expressed as

$$I = -\log p.$$

However, this formulation already presupposes the existence of a state space.

In other words, a set of possible states must already exist before information can even be defined.

The present theory rejects this ordering.

Information is not a state.

Information is not a difference between states.

Instead, information is the structure that remains preserved after the action of ontological constraints.

4.3 Information as a Preservation Structure

We therefore define information as

$$\boxed{\text{Information} = \text{Preserved Structure.}}$$

The crucial point is that preservation is not equivalent to fixation.
 Preservation means maintaining ontological consistency.
 Accordingly, the Future Information Field is not a completely fixed universe.
 Rather, it is a preservation structure that remains capable of manifestation.
 For this reason, the Future Information Field is not a static object.
 Instead, it intrinsically contains the operation of preservation itself.

4.4 The Preservation Operator

Let

$$\mathcal{P}$$

denote the preservation operator.
 It is defined by

$$\mathcal{P} : F \rightarrow F,$$

such that

$$\mathcal{P}(F) = F.$$

This expression signifies that the Future Information Field possesses a self-preserving structure.

Importantly, this is not an equation of temporal evolution.

Time has not yet emerged.

Preservation here is not temporal persistence but ontological consistency.

4.5 The Future Information Tensor

To describe the local structure of the Future Information Field, this paper introduces the Future Information Tensor.

Let

$$\mathcal{F}_{AB}$$

denote the Future Information Tensor.
 The indices

$$A, B$$

are not spacetime coordinates.

Rather, they represent ontological degrees of preservation.

The Future Information Tensor describes the consistency relations among preservation structures.

Therefore,

$$\mathcal{F}_{AB} \neq g_{\mu\nu},$$

and likewise,

$$\mathcal{F}_{AB} \neq T_{\mu\nu}.$$

The Future Information Tensor is therefore a preservation tensor that precedes every space-time tensor.

4.6 The Future Information Metric

Preservation structures possess not spatial distance but ontological consistency.

Define

$$\Gamma(A, B)$$

as the preservation consistency between two preservation structures.

Its range satisfies

$$0 \leq \Gamma \leq 1.$$

The limiting cases are

$$\Gamma = 1$$

for complete preservation,
and

$$\Gamma = 0$$

for total loss of consistency.

The Future Information Tensor satisfies the preservation condition

$$\nabla_{\Omega} \mathcal{F}_{AB} = 0,$$

where

$$\nabla_{\Omega}$$

is the ontological preservation connection.

Unlike the covariant derivative in differential geometry,
this connection is defined on preservation structures themselves.

4.7 Unification with the Meaning Density Tensor

Paper 91 introduced the Meaning Density Tensor as the mathematical description of projections from high-dimensional preservation structures into analytical language.

Within the present framework, the Meaning Density Tensor

$$M_{AB}$$

is interpreted as a local projection of the Future Information Tensor.

Thus,

$$M_{AB} = \Pi_M(\mathcal{F}_{AB}).$$

Here,

$$\Pi_M$$

is the Meaning Projection Operator.

Meaning therefore possesses no independent ontological existence.

Instead, meaning is the projection of the Future Information Tensor into cognitive space.

4.8 Unification with the Constraint Tensor

Ontological constraints may likewise be represented tensorially.

Define

$$C_{AB}$$

as the Constraint Tensor.

This tensor introduces conditions of impossibility among preservation structures.

Consequently,

the Future Information Tensor is transformed according to

$$\mathcal{F}'_{AB} = C_{AB} \mathcal{F}_{AB}.$$

Here,

$$\mathcal{F}'_{AB}$$

denotes the preservation tensor after the action of ontological constraints.

4.9 The Premise Update Tensor

Within Premise Update Theory,

updating is not interpreted as merely adding knowledge.

Instead,

it represents an update of the preservation structure itself.

Define the Premise Update Tensor

$$U_{AB}$$

by

$$U_{AB} : \mathcal{F}_{AB} \rightarrow \mathcal{F}_{AB}^*.$$

Importantly,

updating does not correspond to adding information.

Rather,

it corresponds to a rearrangement of the configuration of ontological constraints.

Thus,

updating the Future Information Tensor means modifying the arrangement of constraints while preserving the underlying preservation structure.

4.10 The Tensor Unification Equation

The fundamental tensors of Future Convergence Theory are therefore

$$\{\mathcal{F}_{AB}, C_{AB}, M_{AB}, U_{AB}\}.$$

These tensors are not independent objects.

Instead,

their relationships are expressed by

$$U_{AB} = \Phi(C_{AB}, \mathcal{F}_{AB}).$$

Furthermore,

the Meaning Density Tensor is obtained through

$$M_{AB} = \Pi_M(U_{AB}).$$

Meaning is therefore nothing more than the cognitive projection of the Future Information Tensor.

4.11 Projection into Cosmology

Let

$$P_X$$

denote Future Projection into cosmological manifestation.

Then the Future Information Tensor is projected into the cosmological tensor

$$T_{\mu\nu} = P_X(\mathcal{F}_{AB}).$$

Importantly,

$$T_{\mu\nu}$$

is not identified directly with the conventional energy-momentum tensor.

Instead,

it represents the cosmological projection of the Future Information Tensor.

Accordingly,

gravity describes not the preservation structure itself but the manifestation of that preservation structure after projection into three-dimensional spacetime.

4.12 Conclusion of This Chapter

This chapter has redefined the Future Information Field as an ontological preservation structure.

Furthermore,

the Future Information Tensor

$$\mathcal{F}_{AB}$$

has been introduced and unified mathematically with the Meaning Density Tensor, the Constraint Tensor, and the Premise Update Tensor.

As a consequence,

cognitive theory,

meaning theory,

and cosmology are no longer independent theoretical frameworks.

Instead,

they become distinct projections of a single preservation tensor.

The next chapter redefines the Quantum Allowance Layer not as a space of possibilities but as an ontological margin of indeterminacy and derives the necessary and sufficient conditions under which Future Projection becomes possible.

5 Redefinition of the Quantum Allowance Layer — The Quantum Layer as an Ontological Margin of Indeterminacy

5.1 Purpose of This Chapter

In Paper 93, the Quantum Allowance Layer was introduced as an intermediate layer situated between the Future Information Field and the three-dimensional Manifestation Layer.

However, the ontological question

What exactly does quantum allowance allow?

remained insufficiently defined.

In conventional quantum theory, quantum states are described in terms of superposition, probability amplitudes, wave functions, and wave-function collapse induced by observation.

All of these descriptions, however, presuppose that a state space already exists.

The present theory rejects this assumption.

Quantum is neither a state nor a possibility.

This chapter therefore redefines the Quantum Allowance Layer as an ontological margin of indeterminacy.

5.2 The Necessity of an Ontological Margin of Indeterminacy

The previous chapters established that the Future Information Field is a preservation structure generated through ontological constraints.

Nevertheless, preservation alone cannot produce manifestation.

The reason is straightforward.

A perfectly preserved structure cannot change.

A structure incapable of change cannot manifest.

Accordingly, an additional ontological condition must exist between preservation and manifestation.

This paper refers to this condition as the *Margin of Indeterminacy*.

5.3 Definition of the Margin of Indeterminacy

Definition The Margin of Indeterminacy is the degree of ontological freedom that remains unfixed while preserving ontological consistency.

It is denoted by

$$W.$$

Importantly,

$$W$$

does not represent the number of states,
nor degrees of freedom in the conventional physical sense.

Rather,

the Margin of Indeterminacy measures the impossibility of complete fixation.

Therefore,

$$W \neq \text{Entropy},$$

and

$$W \neq \text{Probability}.$$

5.4 Definition of the Quantum Allowance Layer

Let

$$Q$$

denote the Quantum Allowance Layer.

It is defined by

$$Q = \{x \in F \mid 0 < W(x) < W_{\max}\}.$$

Thus,

the Quantum Allowance Layer is the region of the Future Information Field that preserves an ontological margin of indeterminacy.

Neither complete fixation nor complete freedom is permitted within this layer.

5.5 Axioms of the Quantum Layer

The Quantum Allowance Layer satisfies the following axioms.

Axiom Q1 (Prohibition of Complete Fixation) Complete fixation is impossible.

$$W > 0.$$

Axiom Q2 (Prohibition of Complete Freedom) Complete freedom is likewise impossible.

$$W < W_{\max}.$$

Axiom Q3 (Preservation of the Underlying Structure) The preservation structure remains unchanged.

$$\mathcal{P}(F) = F.$$

Axiom Q4 (Collapse of the Margin after Projection) The Margin of Indeterminacy collapses after Projection.

$$W \rightarrow 0.$$

This process differs fundamentally from wave-function collapse.

Within the present theory,

it is referred to as *Ontological Fixation through Projection*.

5.6 What Does Quantum Actually Do?

According to this theory,

quantum allowance does not generate possibilities.

It does not select states.

It does not determine the future.

Its role is uniquely defined as

postponing ontological fixation.

Quantum therefore delays manifestation.

This postponement enables Projection to occur while preserving ontological consistency.

5.7 Conditions for Projection

Let

$$P_X$$

denote Projection.

Projection occurs only if

$$W < W_c,$$

where

$$W_c$$

is the critical Margin of Indeterminacy.

Once the margin falls below this critical value,

Projection begins.

Conversely,

if

$$W > W_c,$$

manifestation cannot occur.

Projection therefore requires no observer.

It is initiated automatically once the Margin of Indeterminacy reaches its ontological critical value.

5.8 Future Projection

Future Projection is defined as the mapping

$$P_X : Q \rightarrow X,$$

where

$$X$$

denotes the three-dimensional Manifestation Layer.

Projection is not a generative operation.

It does not create states.

Instead,

it maps preservation structures into manifested structures.

Projection is therefore an operation of fixation rather than generation.

5.9 Unification with the Quantum Creative Margin

Previous studies treated the Quantum Creative Margin as the space preserving creative possibilities.

The present theory redefines it as the Margin of Indeterminacy.

Creation becomes possible only because a finite unfixed margin exists.

If

$$W = 0,$$

the universe becomes completely fixed.

Conversely,
if

$$W \rightarrow \infty,$$

no stable structure can emerge.
Thus,
creation is guaranteed by the finiteness of the Margin of Indeterminacy.

5.10 The Relationship between Quantum Allowance and Gravity

Within the present theory,
Quantum Allowance and Gravity are distinguished by their ontological roles.
Quantum Allowance preserves the Margin of Indeterminacy.
Gravity eliminates it through ontological fixation.
They are therefore not competing theories.
Rather,
they describe successive stages of the same manifestation process.
The ontological sequence becomes

$$F \longrightarrow Q \longrightarrow G \longrightarrow X,$$

where

$$G$$

denotes the operation of ontological fixation.

5.11 Relationship to Dark Matter

A portion of the Future Information Field never participates in Projection.
Nevertheless,
its preservation structure continues contributing to gravitational constraint.
This paper refers to such regions as *Locally Unfixed Constraints*.
These provide the ontological foundation of dark matter.
Dark matter is therefore not matter that failed to become visible.
Rather,
it consists of preservation structures that never completed Projection while nevertheless contributing to the geometry of spacetime through local constraints.

5.12 Conclusion of This Chapter

This chapter has redefined the Quantum Allowance Layer not as a layer of possibilities but as an ontological Margin of Indeterminacy.

Consequently,
quantum phenomena are interpreted neither as probability nor as possibility.
Instead,
they constitute the unfixed preservation layer through which the Future Information Field maintains ontological consistency while awaiting Projection.

Furthermore,
Projection has been formulated as a structural necessity that occurs automatically once the Margin of Indeterminacy reaches its critical value.

The next chapter redefines Gravity not as a force acting within spacetime but as the ontological fixation operator that establishes the existence of the three-dimensional universe through Future Projection.

6 Redefinition of Gravity — Gravity as the Ontological Fixation Operator

6.1 Purpose of This Chapter

The previous chapter redefined the Quantum Allowance Layer as the ontological Margin of Indeterminacy.

Consequently, quantum phenomena were no longer interpreted as possibility or probability, but as an unfixed preservation structure within the Future Information Field.

However, the Margin of Indeterminacy alone cannot produce a three-dimensional universe.

An unfixed preservation structure remains ontologically preserved, but it has not yet become an observable reality.

Therefore, an additional operation is required to fix an unfixed preservation structure into three-dimensional existence.

This paper defines this operation as *Gravity*.

Gravity is not a force that attracts material bodies.

Rather, gravity is the ontological operation that fixes an unfixed preservation structure into three-dimensional reality.

6.2 Reconsidering the Concept of Gravity

In General Relativity, gravity is represented by spacetime curvature,

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.$$

Within that framework, gravity is understood as a consequence of the distribution of matter.

The present theory adopts a different ontological ordering.

Matter is not primary.

Manifestation is primary.

Accordingly, gravity is redefined not as an effect generated by matter, but as the fixation operator that enables matter itself to become manifested.

6.3 Ontological Fixation

Definition Ontological Fixation is the operation that maps an unfixed preservation structure from a manifestable state into an actual manifested state.

This operation is denoted by

$$\mathcal{G}.$$

Thus,

$$\mathcal{G} : Q \rightarrow X,$$

where

$$Q$$

is the Quantum Allowance Layer,
and

$$X$$

is the three-dimensional Manifestation Layer.

6.4 Axioms of Ontological Fixation

The Ontological Fixation Operator satisfies the following axioms.

Axiom G1 (Conservation of Information) Gravity generates no information.

$$|\mathcal{G}(Q)| = |Q|.$$

The total amount of preserved information remains unchanged under fixation.

Axiom G2 (Reduction of the Margin of Indeterminacy) Gravity reduces the Margin of Indeterminacy.

$$W' < W,$$

where

$$W'$$

denotes the Margin of Indeterminacy after fixation.

Axiom G3 (Preservation of Ontological Consistency) Ontological consistency is preserved throughout fixation.

Thus,

$$\nabla_{\Omega} \mathcal{F}_{AB} = 0$$

continues to hold after fixation.

Axiom G4 (Irreversibility) Ontological fixation is generally irreversible.

Accordingly,

$$\mathcal{G}^{-1}$$

does not generally exist.

Axiom G5 (Formation of Three-Dimensional Geometry) Following fixation, three-dimensional geometry becomes well-defined.

Thus,

$$Q \xrightarrow{\mathcal{G}} X.$$

6.5 Projection as the Consequence of Ontological Fixation

The previous chapter defined Projection as the mapping from preservation structures to manifested structures.

The present chapter further specifies that Projection is the consequence of ontological fixation.

Accordingly,

Projection is no longer regarded as an independent operation.

Instead,

$$P_X = \mathcal{G} \circ Q.$$

Projection therefore arises when a preservation structure possessing a finite Margin of Indeterminacy undergoes ontological fixation.

6.6 The Gravitational Constraint Tensor

To describe Ontological Fixation mathematically, we introduce the Gravitational Constraint Tensor,

$$\mathcal{G}_{AB}.$$

This tensor is not a spacetime curvature tensor.

Instead,

it imposes fixation conditions upon the Future Information Tensor.

Accordingly,

$$\mathcal{F}_{AB}^{(X)} = \mathcal{G}_{AB} \mathcal{F}_{AB}^{(Q)},$$

where

$$\mathcal{F}_{AB}^{(Q)}$$

denotes the preservation tensor within the Quantum Allowance Layer,
and

$$\mathcal{F}_{AB}^{(X)}$$

denotes the preservation tensor after manifestation.

6.7 Correspondence with General Relativity

General Relativity describes only the universe after manifestation has occurred.

Within the present theory,

Einstein's field equations therefore appear as the post-manifestation limit of Ontological Fixation.

Specifically,

$$\mathcal{G}_{AB} \longrightarrow g_{\mu\nu},$$

under which

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

is recovered.

General Relativity therefore does not describe Ontological Fixation itself.

Rather,

it describes the geometry of spacetime after Ontological Fixation has already been completed.

6.8 Gravity and Dark Matter

Paper 93 interpreted dark matter as a local constraint originating from the Future Information Field.

The present theory further refines this interpretation.

Definition A local constraint is a preservation structure whose Ontological Fixation has been completed only partially.

Accordingly,
dark matter satisfies

$$0 < W < W_c,$$

while never achieving complete Projection.

Such regions are not visible matter.

Nevertheless,

they contribute to gravitational constraint.

Dark matter is therefore interpreted not as an unknown particle, but as a locally under-fixed preservation structure.

6.9 Gravity and Dark Energy

Dark energy differs fundamentally from local constraints.

Rather,

it represents the average fixation constraint acting upon the universe as a whole.

Define the cosmological mean constraint by

$$\Lambda_\Omega.$$

Then,

the average fixation rate of the universe is given by

$$\langle \mathcal{G} \rangle = \Lambda_\Omega.$$

Dark energy is therefore not an unknown form of energy.

Instead,

it represents the density of Ontological Fixation averaged over the entire universe.

6.10 The Complementary Roles of Quantum Allowance and Gravity

Within the present framework,

Quantum Allowance and Gravity are complementary rather than contradictory.

Quantum Allowance preserves the Margin of Indeterminacy.

Gravity fixes that margin into manifestation.

The ontological hierarchy therefore becomes

$$\Omega \longrightarrow C_\Omega \longrightarrow F \longrightarrow Q \longrightarrow \mathcal{G} \longrightarrow X.$$

The Quantum Allowance Layer preserves future manifestation.

Gravity fixes future manifestation into reality.

Together,

they constitute successive stages of a single manifestation process.

6.11 The Ontological Meaning of Gravity

Gravity is not merely one of the four fundamental interactions.

Instead,

gravity is the Ontological Fixation Operator that enables manifestation itself.

Without gravity,

matter cannot exist.

Spacetime cannot exist.

Observation cannot exist.
Gravity is therefore not a force that governs the universe.
It is the ontological operation that establishes the universe as reality.

6.12 Conclusion of This Chapter

This chapter has redefined Gravity as the Ontological Fixation Operator.

Gravity acts after the Future Information Field and the Quantum Allowance Layer, fixing unfixed preservation structures into three-dimensional reality.

General Relativity is consequently interpreted as the geometry of spacetime after Ontological Fixation has already been completed, with Einstein's field equations recovered as the post-manifestation limit of the theory.

Furthermore,
dark matter has been reinterpreted as a locally under-fixed preservation structure,
while dark energy has been reformulated as the cosmological mean density of Ontological Fixation.

The next chapter derives the strong, weak, and electromagnetic interactions as independent classes of ontological constraints, thereby unifying the four fundamental interactions as the minimal constraint system required for the existence of a manifest universe.

7 Derivation of the Four Fundamental Interactions — Fundamental Interactions as Classes of Ontologically Forbidden States

7.1 Purpose of This Chapter

The previous chapter redefined gravity as the Ontological Fixation Operator.

Gravity alone, however, is insufficient for the existence of the universe.

If gravity alone were present, the universe would become a single globally fixed structure, but no local structures, interactions, or state transitions could arise.

Accordingly, ontological fixation by itself cannot account for the observed universe.

This chapter derives the four fundamental interactions not as independent fundamental forces, but as distinct classes of ontological constraints that generate forbidden regions within the space of possible manifestation.

The essential point is that the existence of four interactions is not assumed a priori.

Instead, this chapter derives the minimum number of independent ontological constraint classes required for a manifest universe to exist.

7.2 Necessary Conditions for a Manifest Universe

For a three-dimensional universe to exist, at least the following conditions must be satisfied:

- (1) Local structures must exist.
- (2) Local structures must be preserved.
- (3) Local structures must be capable of change.
- (4) Local structures must be capable of interaction.
- (5) The universe must possess global coherence.

Each of these properties is directly observed in the physical universe.

Accordingly, they constitute the phenomena that the present theory must explain.

7.3 Derivation of Ontological Constraint Classes

Ontological constraints determine regions that are incapable of existence.

Instead of asking which structures are allowed, this theory asks which modes of existence must be prohibited for the universe to exist.

The required ontological constraint classes are derived by identifying those forbidden modes.

7.4 First Constraint Class: Prohibition of Complete Fixation

If every structure were completely fixed, no change could occur.

Neither time, quantum phenomena, nor creation would exist.

Therefore, complete fixation must be prohibited.

Define

$$C_Q : \text{Complete Fixation} \longrightarrow \text{Forbidden.}$$

This constraint class manifests as the Quantum Allowance Layer.

Quantum allowance is therefore not the existence of possibilities.

Rather, it is the ontological prohibition of complete fixation.

7.5 Second Constraint Class: Prohibition of Complete Independence

If every structure were completely independent, spacetime itself could not exist.

No common universe could emerge.

Accordingly, complete independence must be prohibited.

Define

$$C_G : \text{Complete Independence} \longrightarrow \text{Forbidden.}$$

This constraint class manifests as Gravity.

Gravity is therefore not a force attracting bodies.

Rather, it is the ontological constraint that prevents complete independence among manifested structures.

7.6 Third Constraint Class: Prohibition of Complete Separation

Suppose every local structure could remain permanently separated from every other.

Then particles, atoms, stars, biological systems, and all higher structures could never form.

Accordingly, complete separation must be prohibited.

Define

$$C_S : \text{Complete Separation} \longrightarrow \text{Forbidden.}$$

This constraint class manifests as the Strong Interaction.

Within the present framework, the strong interaction is interpreted as the ontological constraint responsible for preserving local structural integrity.

7.7 Fourth Constraint Class: Prohibition of Complete Isolation

Even if local structures exist, a universe cannot exist if those structures never influence one another.

Without interaction, neither electromagnetic phenomena, chemistry, biological systems, nor information transfer could emerge.

Accordingly, complete isolation must be prohibited.

Define

C_E : Complete Isolation $\longrightarrow$ Forbidden.

This constraint class manifests as the Electromagnetic Interaction.

7.8 Fifth Constraint Class: Prohibition of Complete Invariance

Suppose every structure remained permanently unchanged.

Then decay, evolution, nuclear reactions, stellar evolution, biological evolution, and civilization itself could never arise.

Accordingly, complete invariance must be prohibited.

Define

C_W : Complete Invariance $\longrightarrow$ Forbidden.

This constraint class manifests as the Weak Interaction.

7.9 The Minimal Constraint Theorem

We now state the principal theorem of this chapter.

Theorem 6.1 (Minimal Constraint Theorem) The existence of a manifest universe requires precisely the following five independent ontological constraint classes:

$$\{C_Q, C_G, C_S, C_E, C_W\}.$$

These five constraint classes constitute a necessary and sufficient set.

Proof Necessity.

If

C_Q

is removed,
complete fixation becomes possible,
eliminating all change.

If

C_G

is removed,
complete independence becomes possible,
preventing the formation of a coherent universe.

If

C_S

is removed,
local structures cannot remain stable.

If

C_E

is removed,
all interaction disappears.

If

$$C_W$$

is removed,
state transitions become impossible.
Therefore, every constraint class is necessary.

Sufficiency.

Together, the five constraint classes guarantee
local structure,
structural preservation,
interaction,
state transitions,
global coherence,
and a finite ontological Margin of Indeterminacy.
Consequently, a manifest universe necessarily emerges.

□

7.10 Redefinition of the Fundamental Interactions

The four fundamental interactions are therefore not interpreted as primitive physical forces.
Instead, they are ontological classes of forbidden states.
They may be summarized as

Quantum	: Prohibition of Complete Fixation,
Gravity	: Prohibition of Complete Independence,
Strong Interaction	: Prohibition of Complete Separation,
Electromagnetic Interaction	: Prohibition of Complete Isolation,
Weak Interaction	: Prohibition of Complete Invariance.

These five ontological constraints together constitute the minimal constraint system required for the Future Information Field to manifest as a three-dimensional universe.

7.11 Relationship to the Standard Model

The present theory does not adopt

$$SU(3), \quad SU(2), \quad U(1)$$

as primitive axioms.

Instead, the Standard Model gauge groups are interpreted as possible implementations of the ontological constraint classes

$$C_S, \quad C_W, \quad C_E.$$

Accordingly, gauge symmetry is not regarded as the origin of physical law.
Rather, it is interpreted as one mathematical realization of deeper ontological constraints.
A complete derivation of the Standard Model from these constraints is left for future work.

7.12 Ontological Interpretation

The fundamental interactions do not exist to make the universe operate.

They exist to eliminate modes of existence that would make manifestation impossible.

The universe therefore does not emerge from unrestricted freedom.

Instead, it emerges as the residual structure that remains after ontologically impossible states have been removed.

The four interactions represent the fundamental classes according to which this elimination occurs.

7.13 Conclusion of This Chapter

This chapter has derived the four fundamental interactions as five fundamental classes of ontological constraints.

Consequently,
Quantum,
Gravity,
the Strong Interaction,
the Electromagnetic Interaction,
and the Weak Interaction
are no longer interpreted as independent forces.

Instead, they constitute the minimal ontological constraint system that makes a manifest universe possible.

The next chapter derives dark matter, dark energy, local constraints, and cosmological mean constraints from this ontological framework, establishing their mathematical correspondence with observational cosmology.

8 Dark Matter, Dark Energy, and Cosmological Constraints

8.1 Purpose of This Chapter

The previous chapter established that the four fundamental interactions are not primitive forces but manifestations of ontological constraint classes.

Those constraints determine which states of existence are forbidden.

However, observations indicate that visible matter alone cannot account for the large-scale behavior of the universe.

Galaxy rotation curves, gravitational lensing, structure formation, and the accelerated expansion of the universe all require additional phenomena.

Conventionally these phenomena are attributed to Dark Matter and Dark Energy.

Within the present theory, neither Dark Matter nor Dark Energy is regarded as an independent substance.

Instead, both arise naturally from the global structure of the Future Information Field and its ontological constraint geometry.

8.2 Observable and Non-Observable Constraint Contributions

Let the total ontological constraint field be denoted by

$$\mathcal{C}_{\text{total}}.$$

Decompose it into

$$\mathcal{C}_{\text{total}} = \mathcal{C}_{\text{obs}} + \mathcal{C}_{\text{hidden}},$$

where

$$\mathcal{C}_{\text{obs}}$$

represents constraints directly manifested through observable matter,
and

$$\mathcal{C}_{\text{hidden}}$$

represents constraints that influence manifestation without directly appearing as observable matter.

Dark Matter and Dark Energy arise from different geometric properties of

$$\mathcal{C}_{\text{hidden}}.$$

8.3 Derivation of Dark Matter

Consider a region containing visible matter density

$$\rho_v.$$

The observable gravitational fixation operator is

$$\Gamma_{\text{obs}} = \Gamma(\rho_v).$$

However,
the Future Information Field extends beyond observable manifestation.
Consequently,
neighboring hidden ontological constraints also contribute to fixation.
Define

$$\Gamma_{\text{hidden}} = \Gamma(\mathcal{C}_{\text{hidden}}).$$

The effective gravitational fixation therefore becomes

$$\Gamma_{\text{eff}} = \Gamma_{\text{obs}} + \Gamma_{\text{hidden}}.$$

An observer measuring only visible matter interprets

$$\Gamma_{\text{hidden}}$$

as additional unseen mass.

This apparent excess is identified as Dark Matter.

Definition Dark Matter is the effective gravitational contribution generated by hidden ontological constraints that do not directly manifest as observable matter.

Accordingly,

Dark Matter is not a new particle.

It is an observational consequence of incomplete access to the total ontological constraint field.

8.4 Derivation of Dark Energy

The Future Information Field does not merely constrain local structures.

Its global geometry also determines the large-scale evolution of manifestation.

Let

$$\Phi_F$$

denote the global Future Information Field.

Its average curvature induces a cosmological fixation gradient

$$\nabla\Gamma_F.$$

When

$$\nabla\Gamma_F > 0,$$

the preferred manifestation direction expands globally.
The cosmological scale factor therefore satisfies

$$\ddot{a} > 0.$$

An observer interprets this phenomenon as cosmic acceleration.

Definition Dark Energy is the cosmological manifestation of the large-scale fixation gradient of the Future Information Field.

It is therefore not vacuum energy.

Rather, it is a global geometric property of ontological constraint space.

8.5 Unified Interpretation

Dark Matter and Dark Energy originate from different aspects of the same ontological structure.

Specifically,

Dark Matter corresponds to hidden local constraint contributions,

whereas

Dark Energy corresponds to the global curvature of the Future Information Field.

Thus,

$$\boxed{\text{Dark Matter} \neq \text{Matter}}$$

and

$$\boxed{\text{Dark Energy} \neq \text{Energy}}.$$

Instead,

both are manifestations of unseen ontological geometry.

8.6 The Cosmological Constraint Theorem

Theorem 7.1 Let

$$\mathcal{C}_{\text{total}} = \mathcal{C}_{\text{obs}} + \mathcal{C}_{\text{hidden}}.$$

Then

observable cosmology necessarily exhibits

- (1) effective excess gravity (Dark Matter),
- (2) accelerated expansion (Dark Energy),

provided

$$\mathcal{C}_{\text{hidden}} \neq 0.$$

Proof The hidden constraint field contributes additional fixation to local gravitational measurements,

producing an apparent excess mass.

Its large-scale curvature simultaneously modifies the preferred manifestation direction, yielding cosmological acceleration.

Both effects arise directly from the decomposition of

$$\mathcal{C}_{\text{total}}.$$

Therefore,

Dark Matter and Dark Energy are necessary observational consequences of hidden ontological constraints. □

8.7 Relation to General Relativity

General Relativity attributes gravitational curvature solely to observable stress-energy.

The present framework extends this description.

The effective Einstein equation becomes

$$G_{\mu\nu} = 8\pi G \left(T_{\mu\nu} + T_{\mu\nu}^{\text{hidden}} \right),$$

where

$$T_{\mu\nu}^{\text{hidden}}$$

is not interpreted as physical matter,

but as the effective stress generated by hidden ontological constraints.

Thus,

the classical Einstein equation appears as the observable limit of a deeper ontological description.

8.8 Observational Consistency

This interpretation is consistent with the principal cosmological observations:

- Galaxy rotation curves,
- Gravitational lensing,
- Large-scale structure formation,
- Cosmic microwave background anisotropy,
- Accelerated cosmic expansion.

The present theory does not require introducing additional invisible particles or unknown vacuum energy.

Instead,

these observations arise from the hidden geometry of ontological constraint space.

8.9 Ontological Interpretation

Visible matter represents only the manifested portion of the Future Information Field.

The remaining hidden constraint structure continues to influence manifestation even though it does not itself become observable.

Accordingly,

Dark Matter and Dark Energy are not missing components of the universe.

Rather,

they are signatures of the incompleteness of observation relative to the full ontological constraint geometry.

8.10 Conclusion of This Chapter

This chapter has shown that Dark Matter and Dark Energy emerge naturally from the decomposition of the ontological constraint field.

Dark Matter corresponds to hidden local fixation contributions,
whereas Dark Energy corresponds to the global fixation gradient of the Future Information Field.

Neither requires the introduction of new physical substances.

Instead,

both are observational manifestations of the deeper ontological geometry developed throughout this theory.

The next chapter derives quantum measurement, wavefunction collapse, and probabilistic observation from the Margin of Indeterminacy, completing the bridge between ontology and quantum mechanics.

9 Projection Asymmetry, Baryon Asymmetry, and CP Violation

9.1 Purpose of This Chapter

The previous chapter demonstrated that Dark Matter and Dark Energy emerge naturally from the hidden geometry of ontological constraints.

The present chapter addresses another fundamental problem in modern physics:

Why is the observable universe dominated by matter rather than antimatter?

Within the Standard Model, baryogenesis requires three conditions proposed by Sakharov:

- (1) Baryon number violation,
- (2) CP violation,
- (3) Departure from thermal equilibrium.

Although these conditions explain how an asymmetry may evolve, they do not explain why asymmetry exists at all.

Future Convergence Theory proposes that asymmetry originates prior to particle physics.

Matter dominance is not generated after manifestation.

Instead, it is inherited from the ontological asymmetry of Projection itself.

9.2 Projection Is Fundamentally Non-Symmetric

Projection is the mapping

$$P_X : F \rightarrow X.$$

If Projection were perfectly symmetric,

every manifestation would possess an equally probable mirror manifestation.

Consequently,

$$N_{\text{matter}} = N_{\text{antimatter}}.$$

The observable universe contradicts this expectation.
Therefore,
Projection cannot be fundamentally symmetric.

9.3 Projection Asymmetry Tensor

Define the Projection operator

$$P_X.$$

Its transpose (or symmetry-conjugate operator) is denoted by

$$P_X^T.$$

The Projection Asymmetry Tensor is defined as

$$A_{AB} = P_X - P_X^T.$$

If

$$A_{AB} = 0,$$

Projection is perfectly symmetric.
Future Convergence Theory instead assumes

$$\boxed{A_{AB} \neq 0.}$$

This asymmetry exists before particles,
before spacetime,
and before conventional physical interactions.

9.4 Manifestation Preference

Projection assigns realization weights to reachable structures.

Let

$$R_M$$

denote the realization rate of matter configurations,
and

$$R_A$$

the realization rate of antimatter configurations.
Projection asymmetry implies

$$R_M \neq R_A.$$

The observable universe corresponds to the branch satisfying

$$R_M > R_A.$$

Matter dominance therefore arises directly from ontological manifestation.

9.5 Baryon Asymmetry

Let

$$B$$

denote baryon number,
and

$$\bar{B}$$

the antibaryon number.
The baryon asymmetry is

$$\Delta B = B - \bar{B}.$$

Rather than emerging from particle interactions,
Future Convergence Theory proposes

$$\Delta B = f(A_{AB}),$$

where

$$f$$

maps Projection asymmetry into observable baryon imbalance.
Therefore,
baryon asymmetry is the cosmological manifestation of ontological Projection asymmetry.

9.6 Origin of CP Violation

Within conventional particle physics,
CP violation is regarded as one of the causes of baryogenesis.
The logical order is reversed in Future Convergence Theory.
Projection asymmetry is primary,
whereas CP violation is its manifestation within particle physics.
Accordingly,

$\text{Projection Asymmetry} \implies \text{CP Violation.}$

CP violation is therefore interpreted as the particle-level expression of deeper ontological geometry.

9.7 Local Structure Requires Asymmetry

Perfect symmetry implies zero structural preference.

Without preference,
no stable local structures can emerge.
Consequently,
galaxies,
stars,
atoms,
chemistry,
and life would all fail to develop.
Projection asymmetry is therefore not an accidental imperfection.
It is a necessary condition for structured manifestation.

9.8 Early-Universe Interpretation

During the earliest stage of manifestation,
Projection proceeds rapidly.
Let

$$R(t)$$

denote the realization rate.
Initially,

$$\frac{dR}{dt} \gg 0.$$

The asymmetry tensor

$$A_{AB}$$

therefore determines the matter-antimatter imbalance at the earliest stage of cosmic manifestation.

The observed baryon asymmetry is preserved as the universe evolves.

9.9 The Projection Asymmetry Theorem

Theorem 8.1 If

$$A_{AB} \neq 0,$$

then

$$R_M \neq R_A,$$

and consequently

$$\Delta B \neq 0.$$

Proof The Projection operator assigns unequal realization weights to mutually conjugate configurations.

Since baryon and antibaryon states correspond to distinct Projection branches,
unequal realization probabilities imply unequal baryon abundance.

Therefore,

Projection asymmetry necessarily produces baryon asymmetry.

□

9.10 Observational Predictions

The present theory predicts that

- (1) CP violation ultimately reflects ontological geometry,
- (2) baryon asymmetry is determined before particle interactions,
- (3) different Projection geometries imply different matter dominance,
- (4) future unified theories should derive CP violation from Projection rather than assume it.

These predictions distinguish Future Convergence Theory from conventional baryogenesis models.

9.11 Relation to Existing Physics

Future Convergence Theory does not reject the Sakharov conditions.

Instead,

it interprets them as effective descriptions operating after Projection.

The ontological sequence becomes

Projection Asymmetry $\rightarrow$ CP Violation $\rightarrow$ Baryogenesis $\rightarrow$ Matter-Dominated Universe.

Thus,

particle physics describes the consequences of ontological asymmetry,

rather than its origin.

9.12 Conclusion of This Chapter

This chapter has shown that baryon asymmetry and CP violation arise naturally from the intrinsic asymmetry of Projection.

Matter dominance is therefore not an accidental consequence of particle interactions,

but a necessary manifestation of the ontological structure of Future Convergence Theory.

The next chapter applies the same framework to the origin of the early universe and cosmic inflation, interpreting inflation as the rapid activation of ontological constraints rather than as the dynamics of an independent inflaton field.

10 The Early Universe and Cosmic Inflation: Temporal Evolution of Ontological Constraints

10.1 Purpose of This Chapter

The previous chapter derived baryon asymmetry and CP violation from the intrinsic asymmetry of Projection.

The remaining question concerns the origin of cosmic manifestation itself.

In contemporary cosmology, cosmic inflation is typically explained by introducing an inflaton field whose potential drives an epoch of exponential expansion.

Although inflation successfully explains the observed flatness, homogeneity, and isotropy of the universe, the physical origin of the inflaton remains unknown.

Future Convergence Theory adopts a different approach.

Inflation is not interpreted as the dynamics of an additional scalar field.

Instead, it is understood as the rapid activation of ontological constraints within the Future Information Field.

Accordingly, the birth of the universe corresponds not to the creation of spacetime from nothing, but to the initiation of ontological manifestation through constraint activation.

10.2 The Pre-Manifestation State

Prior to manifestation, neither spacetime nor matter exists.

Only the undifferentiated preservation structure

Ω

is defined.

At this stage,

t

is not a physical time coordinate.

Instead,

the ordering of events is ontological rather than temporal.

Consequently,

phrases such as "before the universe" refer to logical order rather than chronological sequence.

10.3 Activation of Ontological Constraints

The first irreversible event is the action of the ontological constraint set

$$C_{\Omega}.$$

The undifferentiated preservation structure is partitioned into

$$\Omega \longrightarrow \Omega_{\text{reachable}} \cup \Omega_{\text{forbidden}}.$$

The reachable component is subsequently preserved as the Future Information Field,

$$F = \Phi(\Omega_{\text{reachable}}).$$

Thus,

the origin of the universe is identified with the activation of ontological constraints rather than with the emergence of physical spacetime.

10.4 Constraint Density

Define the local ontological constraint density as

$$\lambda.$$

Unlike ordinary matter density,

$$\lambda$$

measures the degree of ontological fixation.

Initially,

constraint activation proceeds extremely rapidly,
so that

$$\frac{d\lambda}{dt} \gg 0.$$

Here,

$$t$$

denotes ontological ordering rather than physical time.

This rapid increase marks the beginning of manifestation.

10.5 Inflation as Constraint Activation

Future Convergence Theory defines inflation through the behavior of the ontological constraint density.

Let

$$\lambda_c$$

denote the critical constraint density.

Inflation begins when

$$\frac{d\lambda}{dt} > \lambda_c.$$

At this stage,

Projection expands rapidly throughout the Future Information Field,

allowing an enormous portion of the reachable structure to become manifested within a very short ontological interval.

To an observer inside spacetime,

this process appears as exponential cosmic expansion.

Therefore,

inflation is interpreted as rapid ontological manifestation rather than rapid spatial expansion.

10.6 Relation to the Inflaton

The inflaton field is not adopted as a fundamental axiom.

Instead,

its effective behavior is interpreted as the macroscopic description of the rapidly changing ontological constraint density,

$$\frac{d\lambda}{dt}.$$

Thus,

the inflaton becomes an effective approximation of deeper ontological dynamics rather than an elementary physical field.

10.7 Homogeneity of the Universe

The observed homogeneity of the universe follows naturally.

Since the Future Information Field exists as a single preservation structure prior to Projection,

constraint activation propagates throughout the entire reachable structure.

Large-scale uniformity therefore emerges automatically,

without requiring distant spacetime regions to establish causal contact after manifestation.

10.8 Flatness Problem

The observed spatial flatness of the universe is interpreted geometrically.

When the constraint density approaches spatial uniformity,

$$\nabla\lambda \rightarrow 0,$$

Projection produces a nearly uniform manifestation geometry.

Consequently,

the resulting spacetime appears spatially flat.

Flatness is therefore interpreted as a manifestation of uniform ontological constraint density.

10.9 Horizon Problem

In the conventional picture,

widely separated regions of the cosmic microwave background possess nearly identical temperatures despite having no causal contact.

Future Convergence Theory resolves this naturally.

Before Projection,

all reachable structures belong to the same Future Information Field.

Therefore,

they already possess ontological consistency before spacetime emerges.

The apparent horizon problem is therefore not fundamental,

but a consequence of interpreting pre-manifestation structures solely through spacetime geometry.

10.10 Cosmological Constant

The cosmological constant

$$\Lambda$$

is reinterpreted as the equilibrium value of the average ontological constraint density,

$$\Lambda = \Lambda_{\Omega} = \langle \lambda \rangle.$$

Accordingly,

the cosmological constant is not vacuum energy.

Instead,

it measures the average fixation density of the manifested universe.

10.11 Primordial Fluctuations

The fluctuations observed in the Cosmic Microwave Background are interpreted differently.

Rather than arising solely from quantum fluctuations,

they correspond to local variations of ontological constraint density,

$$\delta\lambda(x).$$

Projection transfers these fluctuations into spacetime,

where they later seed galaxy formation.

Thus,

primordial structure reflects the geometry of ontological constraints.

10.12 Inflation Theorem

Theorem 9.1 Suppose that

$$\frac{d\lambda}{dt} > \lambda_c.$$

Then Projection propagates rapidly throughout the reachable Future Information Field,

producing a manifested universe with large-scale homogeneity,

near-flat geometry,

and coherent primordial structure.

Proof Rapid activation of ontological constraints increases the realized portion of the Future Information Field over an extremely short ontological interval.

Because Projection acts upon a unified preservation structure,
all manifested regions inherit global consistency.

Consequently,

homogeneity,

flatness,

and coherent primordial fluctuations emerge simultaneously.

□

10.13 Observational Predictions

Future Convergence Theory predicts that

- (1) inflation is governed by constraint activation rather than an elementary inflaton field,
- (2) the cosmological constant reflects average ontological fixation density,
- (3) primordial fluctuations originate from variations in ontological constraints,
- (4) future CMB observations may reveal signatures inconsistent with purely quantum-origin fluctuations.

These predictions provide possible observational tests distinguishing this framework from conventional inflationary cosmology.

10.14 Conclusion of This Chapter

This chapter has reinterpreted the origin of the universe and cosmic inflation as the temporal evolution of ontological constraints.

Inflation is understood as the rapid activation of the Future Information Field rather than as the dynamics of a separate inflaton field.

The flatness problem,

the horizon problem,

the cosmological constant,

and primordial fluctuations are all explained within a unified ontological framework.

The next chapter integrates the Future Information Tensor, the Semantic Density Tensor, the Constraint Tensor, and the Premise Update Tensor into a single unified preservation tensor theory.

11 Unified Preservation Tensor Equation: Integrating Cognition and Cosmology

11.1 Purpose of This Chapter

The preceding chapters established the ontological hierarchy

$$\Omega \rightarrow C_\Omega \rightarrow F \rightarrow Q \rightarrow \mathcal{G} \rightarrow P_X \rightarrow X.$$

This hierarchy explains the emergence of observable reality from an undifferentiated preservation structure through ontological constraints.

However, Future Convergence Theory is intended to be more than a theory of cosmology.

The same mathematical structure is proposed to underlie cognition, language, semantics, physical reality, and all forms of manifestation.

The purpose of this chapter is therefore to unify the mathematical objects introduced throughout this work—

$$\mathcal{F}_{AB}, \quad C_{AB}, \quad M_{AB}, \quad U_{AB},$$

into a single preservation tensor framework.

The central claim of this chapter is that cognition and cosmology are not independent domains.

Rather, they are distinct projections of the same underlying preservation tensor.

11.2 The Preservation Tensor

Define the Future Information Tensor

$$\mathcal{F}_{AB}$$

as the fundamental preservation tensor.

Unlike conventional tensors,

$$\mathcal{F}_{AB}$$

does not describe spacetime,

matter,

or energy.

Instead,

it represents the preserved ontological consistency of the reachable structure.

Consequently,

$$\mathcal{F}_{AB}$$

exists prior to spacetime,

prior to physical geometry,

and prior to observation.

11.3 Preservation Connection

Introduce the preservation connection

$$\nabla_{\Omega}.$$

The defining property of the preservation tensor is

$$\nabla_{\Omega}\mathcal{F}_{AB} = 0.$$

This equation is not a spacetime covariant derivative.

Instead,

it expresses ontological consistency.

The preservation structure remains internally coherent throughout every stage of manifestation.

11.4 Projection Operators

The preservation tensor itself is never observed directly.

Observable phenomena arise through projection operators.

Define

$$\Pi_i : \mathcal{F} \rightarrow R_i,$$

where

$$R_i$$

denotes an arbitrary manifestation domain.

Different projection operators correspond to different observable realities.

The underlying preservation tensor remains identical.

11.5 Cognitive Projection

Let

$$\Pi_C$$

denote the cognitive projection.

The Semantic Density Tensor is then defined as

$$M_{AB} = \Pi_C(\mathcal{F}_{AB}).$$

Meaning is therefore not fundamental.

Meaning is the cognitive projection of the preservation tensor.

Semantic structure is thus derived rather than assumed.

11.6 Linguistic Projection

Language is itself a secondary projection.

Let

$$\Pi_L$$

denote the linguistic projection.

Natural language is obtained through

$$L = \Pi_L(M_{AB}).$$

Therefore,

language does not directly express preservation.

Instead,

it expresses semantic structure,

which itself expresses preservation.

The hierarchy becomes

$$\mathcal{F} \rightarrow M \rightarrow L.$$

11.7 Cosmological Projection

The cosmological projection is denoted by

$$\Pi_U.$$

The corresponding cosmological tensor becomes

$$T_{\mu\nu} = \Pi_U(\mathcal{F}_{AB}).$$

Thus,

the stress-energy tensor is not fundamental.

It is the spacetime projection of the preservation tensor.

General Relativity therefore describes only one manifestation layer of the deeper ontological structure.

11.8 Constraint Tensor

The ontological constraint tensor is defined by

$$C_{AB}.$$

Rather than generating new structures,
the constraint tensor restricts manifestation by removing forbidden regions.
The constrained preservation tensor becomes

$$\mathcal{F}_{AB}^* = C_{AB} \mathcal{F}_{AB}.$$

Constraint therefore acts through structural restriction rather than information creation.

11.9 Premise Update Tensor

Premise updating is also reformulated.

Define the Premise Update Tensor

$$U_{AB}$$

through

$$U_{AB} : \mathcal{F}_{AB} \rightarrow \mathcal{F}'_{AB}.$$

Importantly,
the update operation does not modify preservation itself.
Instead,
it modifies the configuration of ontological constraints,
thereby altering subsequent projections.

Knowledge acquisition is therefore interpreted as the reconfiguration of constraints rather than the accumulation of information.

11.10 Unified Preservation Equation

The preservation tensor,

constraint tensor,

and update tensor combine into the unified preservation equation

$$\boxed{\mathcal{F}_{AB}^{(n+1)} = U_{AB} \left(C_{AB} \mathcal{F}_{AB}^{(n)} \right).}$$

Here,

$$n$$

does not represent physical time.
It denotes successive stages of ontological updating.
This equation describes preservation,
constraint,
and updating within a single mathematical operation.

11.11 Universal Projection Equation

Every manifestation domain is obtained through projection.

Accordingly,

$$R_i = \Pi_i(\mathcal{F}).$$

The symbol

$$R_i$$

may represent

- physical reality,
- cognition,
- language,
- mathematics,
- biology,
- artificial intelligence,
- social systems,

or any other manifestation domain.

The only distinction lies in the projection operator.

The preservation tensor itself remains unchanged.

11.12 Non-Commutativity of Projection

Different projections generally do not commute.

That is,

$$\Pi_A \Pi_B \neq \Pi_B \Pi_A.$$

Consequently,

the order of projection influences the resulting manifestation.

This non-commutativity applies not only to quantum measurement,
but also to semantic interpretation,

cognitive reasoning,

scientific observation,

and physical manifestation.

Projection order is therefore a universal structural property.

11.13 Unified Ontological Structure

The complete ontological hierarchy may now be written as

$$\Omega \rightarrow C \rightarrow \mathcal{F} \rightarrow U \rightarrow \Pi_i \rightarrow R_i.$$

Every observable domain is generated through this sequence.

Cosmology,

cognition,

semantics,

language,

and physics therefore share the same mathematical foundation.

They differ only in their projection operators.

11.14 Ontological Interpretation

Future Convergence Theory does not begin with matter,

energy,

or spacetime.

Neither does it begin with information or meaning.

Instead,

the universe begins with preservation.

Ontological constraints determine reachable structure.

Updating reorganizes those constraints.

Projection transforms preservation into observable manifestation.

Accordingly,

reality itself is understood as an ongoing sequence of projections applied to an underlying preservation structure.

11.15 Conclusion of This Chapter

This chapter has unified the Future Information Tensor,

the Semantic Density Tensor,

the Constraint Tensor,

and the Premise Update Tensor within a single preservation tensor framework.

The resulting theory provides a common mathematical foundation for cosmology,

cognition,

semantics,

language,

and physical reality.

The next chapter derives the empirical predictions and falsifiability conditions of Future Convergence Theory, demonstrating that the framework is not merely philosophical but scientifically testable.

12 Falsifiability and Empirical Predictions of Future Convergence Theory

12.1 Purpose of This Chapter

A scientific theory must not only explain existing observations but also produce predictions that can, in principle, be falsified.

Future Convergence Theory (FCT) is proposed as a mathematical theory of ontological manifestation rather than a purely philosophical framework.

Accordingly, its validity depends upon empirical consequences that distinguish it from existing physical theories.

This chapter summarizes the principal observational predictions of the theory and specifies the conditions under which it would be rejected.

12.2 Principle of Scientific Falsifiability

The explanatory power of a theory alone is insufficient.

A theory becomes scientific only if observations exist that could demonstrate its failure.

Accordingly,

Prediction $\neq$ Explanation.

The purpose of FCT is therefore not merely to reinterpret existing physics but to generate measurable consequences that differ from competing theories.

12.3 Black Hole Information

Within FCT,

information is preserved by the Future Information Tensor rather than by observable matter.

Consequently,

black-hole evaporation does not destroy information.

Instead,

information remains encoded within the underlying preservation structure.

Therefore,

the theory predicts that black-hole evaporation is fundamentally unitary,

although the preserved information need not reappear as ordinary particles.

Potential Falsification If future observations demonstrate irreversible destruction of physical information,

the preservation principle of FCT would be invalidated.

12.4 The Page Curve

The Page curve is interpreted differently from conventional quantum gravity.

Rather than describing microscopic quantum degrees of freedom,

the Page curve represents the projection of the preservation consistency function

$$\Gamma(t).$$

Different classes of black holes therefore predict slightly different Page curves,

reflecting variations in ontological constraint geometry rather than microscopic particle statistics.

12.5 Gravitational-Wave Ringdown

General Relativity predicts characteristic quasi-normal modes following black-hole mergers.

Future Convergence Theory predicts an additional contribution originating from the relaxation of ontological fixation.

Consequently,

late-time ringdown signals should contain an extremely weak correction associated with the restoration of preservation equilibrium.

Potential Falsification If future gravitational-wave detectors achieve sufficient precision and consistently exclude any such correction,

the fixation dynamics proposed here would require revision.

12.6 Gravitational-Wave Echoes

In this framework,

gravitational-wave echoes do not originate from horizon reflection.

Instead,

they arise from delayed rearrangements of hidden ontological constraints within the Future Information Field.

Therefore,

their characteristic frequencies need not depend exclusively upon horizon geometry.

They should also reflect the local preservation geometry.

12.7 Dark Matter Distribution

Dark Matter is interpreted as hidden ontological fixation rather than unknown particles.

Consequently,

galactic Dark Matter distributions should correlate with the geometry of hidden preservation constraints.

Future observations comparing galaxy formation histories with reconstructed gravitational distributions may therefore distinguish this framework from particle Dark Matter models.

12.8 Dark Energy

Dark Energy corresponds to the average ontological fixation density

$$\Lambda_{\Omega}.$$

Accordingly,

the expansion history

$$H(z)$$

need not exactly coincide with the predictions of the standard

$$\Lambda\text{CDM}$$

model.

Small but systematic deviations at high redshift constitute a possible observational signature.

12.9 The Casimir Effect

The Casimir effect is interpreted not only as a quantum vacuum phenomenon but also as a manifestation of modified ontological boundary constraints.

Accordingly,

Future Convergence Theory predicts extremely small higher-order corrections beyond conventional quantum electrodynamics under carefully controlled boundary geometries.

12.10 The Equivalence Principle

General Relativity predicts exact equality between inertial and gravitational mass.

Within FCT,

this equality corresponds to the limit of complete ontological fixation.

Extremely precise experiments performed under exceptionally large ontological fixation gradients could reveal tiny deviations from exact equivalence.

Potential Falsification Should experimental precision permanently exclude corrections beyond the limits predicted by the theory,

the fixation operator would require modification.

12.11 The Cosmic Microwave Background

Primordial anisotropies are interpreted as projections of fluctuations in ontological constraint density,

$$\delta\lambda.$$

Consequently,

future high-precision measurements of higher-order correlation functions may reveal signatures inconsistent with purely quantum fluctuations.

Detection of such signatures would support the ontological constraint interpretation.

12.12 Inflation

Inflation is interpreted as rapid activation of ontological constraints rather than the evolution of an elementary inflaton field.

Accordingly,

Future Convergence Theory does not require the existence of a fundamental inflaton particle.

Its eventual non-detection would therefore not threaten the present framework.

12.13 Fundamental Interactions

The Standard Model gauge groups

$$SU(3), \quad SU(2), \quad U(1)$$

are interpreted as effective realizations of deeper ontological constraint classes.

Future unification theories are therefore expected to derive gauge symmetry from preservation geometry rather than adopting it as an independent postulate.

12.14 Future Information Tensor

The central prediction of Future Convergence Theory is the existence of the Future Information Tensor,

$$\mathcal{F}_{AB}.$$

This tensor is not directly observable.

Instead,

it contributes systematic corrections across apparently unrelated physical phenomena.

Accordingly,

common mathematical structures should eventually appear in

- black-hole evaporation,
- gravitational waves,
- Dark Matter,
- Dark Energy,
- the Cosmic Microwave Background,
- the Casimir effect,
- quantum measurement.

Observation of consistent correction terms across these phenomena would provide strong evidence for the preservation tensor.

12.15 Global Falsification Criteria

Future Convergence Theory must be rejected or fundamentally revised if any of the following are established:

1. Information is shown to be fundamentally destroyed.
2. A complete unification of quantum mechanics and gravitation is achieved without introducing Projection or preservation structures.

3. Dark Matter is completely explained by directly observed elementary particles.
4. No common mathematical correction associated with the Future Information Tensor appears in any physical observation.
5. A more economical theory explains all currently unresolved problems without requiring ontological constraints.

These criteria ensure that the theory remains scientifically testable.

12.16 Conclusion of This Chapter

This chapter has summarized the principal observational consequences of Future Convergence Theory.

The theory predicts measurable signatures in black-hole physics,
gravitational waves,
Dark Matter,
Dark Energy,
the Cosmic Microwave Background,
the Casimir effect,
and precision gravitational experiments.

It also specifies explicit conditions under which the theory would be rejected.

Future Convergence Theory is therefore presented not as metaphysical speculation but as a falsifiable mathematical framework for fundamental physics.

The next chapter formulates the complete axiomatic system of Future Convergence Theory, providing the minimal mathematical foundation from which the entire framework may be derived.

13 The Axiomatic System of Future Convergence Theory

13.1 Purpose of This Chapter

The previous chapters established the conceptual and mathematical framework of Future Convergence Theory (FCT).

To complete the theory as a formal mathematical system, its fundamental assumptions must be reduced to a minimal set of axioms.

These axioms serve as the foundation from which all subsequent definitions, theorems, and physical predictions are derived.

The objective of this chapter is therefore to formulate the complete axiomatic basis of Future Convergence Theory.

13.2 Principles of Axiom Selection

Every axiom adopted in this theory satisfies the following criteria.

1. It cannot be derived from any preceding axiom.
2. It is consistent with currently established empirical observations.
3. It contains no anthropomorphic assumptions.
4. It does not assume intention, purpose, or teleology.
5. It admits mathematical generalization and further theoretical development.

These requirements ensure that the theory remains both mathematically rigorous and scientifically testable.

13.3 Axiom A1: Undifferentiated Preservation Structure

There exists an undifferentiated preservation structure,

$$\boxed{\exists \Omega.}$$

Within

$$\Omega,$$

none of the following concepts are yet defined:

- space,
- time,
- matter,
- energy,
- probability,
- information,
- meaning.

Only preservation itself exists.

13.4 Axiom A2: Ontological Constraints

A set of ontological constraints

$$C_{\Omega}$$

acts upon

$$\Omega.$$

The constraint operator satisfies

$$C_{\Omega} : \Omega \rightarrow \Omega'.$$

Its role is not to generate new entities,
but to exclude forbidden regions of preservation.

13.5 Axiom A3: Reachable Structure

Following constraint application,
the preservation structure separates into

$$\Omega' = \Omega_R \cup \Omega_F,$$

where

$$\Omega_R$$

denotes the reachable structure,
and

$$\Omega_F$$

denotes the forbidden structure.

Reachability therefore arises as a consequence of constraint rather than as a primitive concept.

13.6 Axiom A4: Future Information Field

The Future Information Field is defined by

$$F = \Phi(\Omega_R).$$

It is not an information storage medium.

Instead,

it is the preservation state of the reachable structure.

13.7 Axiom A5: Quantum Allowance Layer

The Future Information Field does not immediately manifest.

Instead,

it preserves a finite Margin of Indeterminacy,

$$W.$$

The Quantum Allowance Layer is therefore

$$Q = \{x \in F \mid W(x) > 0\}.$$

Quantum behavior represents preserved indeterminacy rather than probabilistic existence.

13.8 Axiom A6: Ontological Fixation

Manifestation occurs through the ontological fixation operator

$$\mathcal{G},$$

defined by

$$\mathcal{G} : Q \rightarrow X,$$

where

$$X$$

is the manifested spacetime domain.

Gravity is therefore interpreted as ontological fixation rather than as a conventional interaction.

13.9 Axiom A7: Projection

Projection maps preservation into observable manifestation.

It is defined as

$$P_X : F \rightarrow X.$$

Projection creates no new preservation.

It merely expresses preservation within an observable domain.

13.10 Axiom A8: Preservation Tensor

The Future Information Tensor

$$\mathcal{F}_{AB}$$

satisfies

$$\nabla_{\Omega}\mathcal{F}_{AB} = 0.$$

This condition expresses ontological consistency rather than spacetime covariance.

13.11 Axiom A9: Universal Projection

Every observable object

$$R_i$$

is obtained through projection,

$$R_i = \Pi_i(\mathcal{F}).$$

Different observable domains correspond only to different projection operators.

13.12 Axiom A10: Constraint Updating

Preservation itself is invariant.

Only the configuration of constraints changes.

Define the update operator

$$U : C \rightarrow C'.$$

Learning,

adaptation,

and evolution therefore correspond to reconfiguration of constraints rather than modification of preservation.

13.13 Axiom A11: Ontological Asymmetry

Projection is fundamentally asymmetric.

Therefore,

$$A_{AB} \neq 0.$$

This asymmetry produces

- local structure,
- CP violation,
- matter dominance,
- cosmic structure formation.

13.14 Axiom A12: Preservation Consistency

Projection never destroys preservation.

For every projection operator,

$$\Pi_i(\mathcal{F}) \subseteq \mathcal{F}.$$

Manifestation therefore changes representation,
not preservation itself.

13.15 Fundamental Ontological Sequence

Combining the twelve axioms yields the fundamental hierarchy of Future Convergence Theory,

$$\boxed{\Omega \rightarrow C_\Omega \rightarrow \Omega_R \rightarrow F \rightarrow Q \rightarrow \mathcal{G} \rightarrow P_X \rightarrow X.}$$

This sequence is universal.

It applies equally to

- cosmology,
- fundamental physics,
- cognition,
- language,
- biology,
- artificial intelligence.

13.16 The Fundamental Equation

The entire theory may be summarized by the single equation

$$\boxed{R_i = \Pi_i [\mathcal{G} (Q (\Phi (C_\Omega(\Omega))))].}$$

Every manifestation is obtained by applying the same ontological sequence to the underlying preservation structure.

13.17 Conclusion of This Chapter

This chapter has formulated the complete axiomatic system of Future Convergence Theory.

The twelve axioms provide a minimal mathematical foundation from which the Future Information Field,

the Quantum Allowance Layer,

Projection,

Gravity,

the Preservation Tensor,

and all subsequent physical consequences may be derived.

The next chapter proves the principal mathematical theorems of the theory and demonstrates that the entire framework forms a logically closed axiomatic system.

14 Principal Theorems of Future Convergence Theory

14.1 Purpose of This Chapter

The previous chapter established the complete axiomatic foundation of Future Convergence Theory.

An axiomatic system alone, however, does not constitute a complete mathematical theory.

A formal theory requires theorems that are logically derived from its axioms.

The purpose of this chapter is therefore to present the principal theorems that follow directly from the twelve axioms introduced previously.

These theorems establish the logical closure of Future Convergence Theory and provide the mathematical basis for all subsequent applications.

14.2 Theorem 1: Existence Theorem

Statement Suppose that the reachable preservation structure is non-empty,

$$\Omega_R \neq \emptyset.$$

Then the Future Information Field necessarily exists.

That is,

$$\boxed{\Omega_R \neq \emptyset \implies \exists F.}$$

Proof By Axiom A2,

ontological constraints partition the undifferentiated preservation structure.

By Axiom A3,

the reachable preservation structure exists.

By Axiom A4,

the reachable preservation structure is preserved as the Future Information Field.

Therefore,

$$\exists F.$$

□

14.3 Theorem 2: Projection Theorem

Statement If the Margin of Indeterminacy satisfies

$$0 < W < W_c,$$

then Projection necessarily occurs.

Equivalently,

$$\boxed{0 < W < W_c \implies \exists P_X.}$$

Proof Axiom A5 guarantees the existence of the Quantum Allowance Layer.

Axiom A6 defines ontological fixation.

Projection is therefore realized whenever finite ontological indeterminacy exists.

□

14.4 Theorem 3: Constraint Completeness Theorem

Statement The five ontological constraint classes

$$\{C_Q, C_G, C_S, C_E, C_W\}$$

form a necessary and sufficient set for stable manifestation.

Proof Removing any one constraint destroys one of the following indispensable properties:

- indeterminacy,
- global consistency,
- local stability,
- interaction,
- state evolution.

Conversely,
their simultaneous presence guarantees the existence of a stable manifested universe. □

14.5 Theorem 4: Tensor Unification Theorem

Statement The Future Information Tensor,
Constraint Tensor,
Semantic Density Tensor,
and Premise Update Tensor are all projections of a single preservation tensor.
That is,
there exists

$$\mathcal{F}$$

such that

$$M = \Pi_M(\mathcal{F}),$$

$$C = \Pi_C(\mathcal{F}),$$

$$U = \Pi_U(\mathcal{F}).$$

Proof Each tensor differs only through its associated projection operator.
The underlying preservation tensor remains identical.
Therefore,
all such tensors represent different manifestations of one ontological object. □

14.6 Theorem 5: Projection Asymmetry Theorem

Statement Every manifested universe containing stable local structures satisfies

$$\boxed{A_{AB} \neq 0.}$$

Proof Perfect symmetry produces no preferred realization direction.
Without realization preference,
stable local structures cannot emerge.
Therefore,
Projection must possess intrinsic asymmetry. □

14.7 Theorem 6: Matter Dominance Theorem

Statement Projection asymmetry implies unequal realization rates for matter and antimatter,

$$A_{AB} \neq 0 \implies R_M \neq R_A.$$

Consequently,
the manifested universe exhibits baryon asymmetry.

Proof Matter and antimatter correspond to distinct realization branches.
Projection asymmetry assigns unequal realization weights to those branches.
Hence,
matter dominance follows directly. □

14.8 Theorem 7: Dark Constraint Theorem

Statement Hidden ontological constraints necessarily contribute observable gravitational effects despite lacking direct manifestation.

Proof Projection need not fully manifest every reachable preservation structure.
The unmanifested remainder still contributes ontological fixation,
which appears observationally as gravitational influence.
This contribution is interpreted as Dark Matter. □

14.9 Theorem 8: Cosmological Constraint Theorem

Statement The cosmological constant equals the average ontological fixation density,

$$\boxed{\Lambda = \Lambda_{\Omega}.$$

Proof The average fixation density determines the global manifestation geometry.
The cosmological constant describes precisely this large-scale geometric property after Projection.
Therefore,
both quantities coincide. □

14.10 Theorem 9: Projection Hierarchy Theorem

Statement The sequence

$$\Omega \rightarrow C \rightarrow F \rightarrow Q \rightarrow \mathcal{G} \rightarrow P_X \rightarrow X$$

cannot be permuted.

Proof Each operator is defined only on the codomain of its predecessor.
Changing the order destroys the mathematical domain of at least one operator.
Hence,
the hierarchy is unique. □

14.11 Theorem 10: Universal Projection Theorem

Statement Every observable object is a projection of the preservation tensor.

Formally,

for every manifestation

$$R_i,$$

there exists a projection operator

$$\Pi_i$$

such that

$$\boxed{R_i = \Pi_i(\mathcal{F}).}$$

Proof This follows directly from Axiom A9.

Different observable domains correspond only to different projection operators,
not to different preservation structures.

□

14.12 Mathematical Closure

The twelve axioms and the ten principal theorems together generate the complete logical structure of Future Convergence Theory.

The theory therefore possesses the following hierarchy:

$$\boxed{\mathcal{A} \implies \mathcal{T} \implies \mathcal{P},}$$

where

$$\mathcal{A}$$

denotes the axioms,

$$\mathcal{T}$$

the derived theorems,

and

$$\mathcal{P}$$

their physical applications.

This establishes Future Convergence Theory as a mathematically closed formal system.

14.13 Conclusion of This Chapter

This chapter has derived the principal mathematical theorems of Future Convergence Theory directly from its axioms.

Together,

the axioms and theorems establish a logically complete framework connecting preservation,
constraint,

Projection,

quantum theory,

gravitation,

cosmology,

cognition,
and semantics.

The final chapter presents the Future Convergence Principle, the fundamental statement that unifies the entire theory into a single ontological law.

15 The Future Convergence Principle: The Fundamental Law of Future Convergence Theory

15.1 Purpose of This Chapter

The preceding chapters have developed Future Convergence Theory from its ontological foundations through its mathematical formalism, physical implications, empirical predictions, axiomatic structure, and principal theorems.

The purpose of this final chapter is to unify every component of the theory into a single governing principle.

Future Convergence Theory is not intended to be a collection of independent hypotheses.

Instead, every mathematical object introduced throughout this work should emerge from one underlying ontological law.

This final law is called the *Future Convergence Principle*.

15.2 The Future Convergence Principle

The central principle of the theory is stated as follows.

Manifestation is the process by which an undifferentiated preservation structure, after being constrained onto

No assumption of intention,
purpose,
teleology,
or intrinsic possibility is required.
The theory assumes only

Ω ,

the preservation structure,
together with

C_Ω ,

the ontological constraints acting upon it.
Everything else follows from their interaction.

15.3 First Consequence

The universe does not originate from unrestricted possibility.

Instead,

the universe originates from preservation.

Ontological constraints eliminate forbidden structures,

leaving only the reachable preservation structure.

Possibility therefore appears only after ontological constraint has acted.

15.4 Second Consequence

The reachable preservation structure becomes the Future Information Field,

$$F.$$

The Future Information Field does not contain information about the future.

Rather,

it is the preserved ontological structure from which future manifestation becomes possible.

Information is therefore redefined as preserved reachable structure.

15.5 Third Consequence

The Future Information Field does not immediately become reality.

Instead,

it preserves a finite Margin of Indeterminacy,

represented by the Quantum Allowance Layer.

Quantum behavior is therefore interpreted as preserved ontological openness rather than probabilistic uncertainty.

15.6 Fourth Consequence

Gravity is not an interaction between masses.

Gravity is the ontological fixation operator that transforms preserved indeterminacy into observable manifestation.

General Relativity therefore describes the geometry that emerges after ontological fixation has already occurred.

15.7 Fifth Consequence

The fundamental interactions are not primitive forces.

Instead,

they are the minimal ontological constraint classes required for stable manifestation.

Their role is not to drive the universe,

but to prevent structurally impossible modes of existence.

15.8 Sixth Consequence

Dark Matter and Dark Energy are not additional physical substances.

They are distinct observational manifestations of hidden ontological constraints.

Dark Matter corresponds to local hidden fixation,

whereas Dark Energy corresponds to the average global fixation geometry of the Future Information Field.

15.9 Seventh Consequence

Projection is fundamentally asymmetric.

This intrinsic asymmetry gives rise to

- matter dominance,
- CP violation,
- large-scale structure,
- primordial cosmological asymmetry.

Observable asymmetry is therefore inherited from ontological geometry rather than generated by later physical processes.

15.10 Eighth Consequence

Cognition,
language,
semantics,
mathematics,
biology,
artificial intelligence,
and physical reality are not independent domains.
They are distinct projections of the same preservation tensor.
Consequently,
all observable systems possess a common mathematical origin.

15.11 Mathematical Form of the Principle

The complete Future Convergence Principle is expressed by the equation

$$R = \Pi [\mathcal{G} (Q (\Phi (C_{\Omega}(\Omega))))] .$$

Here,

Ω

denotes the undifferentiated preservation structure,

C_{Ω}

the ontological constraint operator,

Φ

the Future Information Field,

Q

the Quantum Allowance Layer,

$\mathcal{G}$

the ontological fixation operator,

Π

the projection operator,
and

R

any manifested reality.

Every observable phenomenon arises through this single ontological sequence.

15.12 Scientific Falsifiability

The Future Convergence Principle remains a scientific hypothesis.

It must therefore be rejected or revised if any of the following conditions are established:

1. A more fundamental theory explains all unresolved phenomena without requiring preservation structures.
2. Projection is shown to be unnecessary for a complete unification of quantum mechanics and gravitation.
3. No observational evidence ever supports common correction terms associated with the Future Information Tensor.
4. Hidden ontological constraints prove incapable of explaining the observed behavior of Dark Matter and Dark Energy.

These conditions preserve the empirical character of the theory.

15.13 Position of Future Convergence Theory

Future Convergence Theory does not replace General Relativity,
Quantum Mechanics,
or the Standard Model.

Instead,

those theories are interpreted as effective descriptions of already manifested reality.

Future Convergence Theory addresses a deeper level,

namely,

the ontological process by which manifestation itself becomes possible.

15.14 The Future Convergence Theorem

Theorem For every observable manifestation

$$R,$$

there exist

$$\Omega, \quad C, \quad F, \quad Q, \quad \mathcal{G}, \quad \Pi,$$

such that

$$\boxed{R = \Pi [\mathcal{G} (Q (\Phi (C(\Omega))))] .}$$

Proof The theorem follows directly from

Axioms A1–A12,

together with the Existence Theorem,

Projection Theorem,

Constraint Completeness Theorem,

Tensor Unification Theorem,

Projection Asymmetry Theorem,

and Universal Projection Theorem.

Every manifested object is therefore uniquely generated through the ontological sequence defined by the Future Convergence Principle.

□

15.15 Final Summary

This work has reconstructed Future Convergence Theory as a unified mathematical ontology founded upon preservation, ontological constraint, and projection.

Within this framework,
possibility is no longer fundamental,
information is redefined as preserved reachable structure,
quantum mechanics describes preserved indeterminacy,
gravity represents ontological fixation,
the fundamental interactions become classes of ontological constraints,
Dark Matter and Dark Energy emerge from hidden preservation geometry,
and cognition, language, cosmology, and physics become different projections of a common preservation tensor.

Accordingly,

Future Convergence Theory is not a theory asking

“What is the universe made of?”

Instead,

it asks

“By what ontological process does the universe become manifest?”

The Future Convergence Principle presented in this chapter provides the unified mathematical answer proposed throughout this work.

A Appendix A: Mathematical Formalization: Conservation Tensor Action Principle and Constraint Operators

A.1 Purpose of this Appendix

This appendix provides the mathematical formalization of the ontological constraint theory constructed in the main text by introducing the conservation tensor space, constraint operators, an action principle, and the corresponding variational field equations. In the main body of this paper, the ontological sequence

$$\Omega \rightarrow C_\Omega \rightarrow \Omega_R \rightarrow F \rightarrow Q \rightarrow \mathcal{G} \rightarrow P_X \rightarrow X$$

was established as the fundamental ontological hierarchy. Based on this sequence, the Future Information Field, the Quantum Allowance Layer, Gravity, Projection, and the fundamental interactions were reconstructed as different manifestations of ontological regions that survive existential constraints. The objective of this appendix is not to replace this ontological sequence directly with physical field equations. Instead, the constraint concept introduced throughout the main text is first formulated rigorously as an operator acting on a conservation tensor space. On this mathematical foundation, an action principle for the Future Information Tensor is introduced, from which the conservation tensor equation, the constraint connection equation, and the unfixed allowance equation are derived. Accordingly, this appendix does not claim that the Einstein equation, the Schrödinger equation, or the Yang–Mills equations have already been completely derived from Future Convergence Theory. Rather, its purpose is to specify the mathematical conditions under which these established physical equations emerge as Projection limits, low-constraint limits, or effective-field approximations of FCT. Under this interpretation, the appendix does not extend the claims of the main text beyond their intended scope. Instead, it provides the formal mathematical framework required to treat ontological constraints, the Future Information Field, the Quantum Allowance Layer, Gravity, and Projection within a consistent analytical formulation.

B Rigorous Definition of Constraint Operators

B.1 Objective

Throughout the main text, ontological constraints were denoted by

$$C_\Omega$$

or

$$C_{AB}.$$

However, at that stage it was intentionally left unspecified whether a constraint should be regarded as a set, a tensor, a contraction, or an operator. In this chapter, constraints are rigorously defined as linear operators acting on the conservation tensor space. This formulation provides a precise mathematical representation of the concepts introduced in the main text, namely impossible regions, reachable regions, and the principle that possibility emerges only after the application of ontological constraints.

B.2 Conservation Tensor Space

Let the conservation tensor on the Future Information Field be denoted by

$$\mathcal{F}_{AB}.$$

The space consisting of all conservation tensors is defined as

$$\mathfrak{F}.$$

Thus,

$$\mathcal{F} \in \mathfrak{F}.$$

The space $\mathfrak{F}$ is equipped with the inner product

$$\langle \mathcal{F}, \mathcal{G} \rangle = \int_{\Omega} \mathcal{F}_{AB} \mathcal{G}^{AB} d\mu_{\Omega},$$

thereby forming the conservation tensor space used throughout this appendix.

B.3 Constraint Operator

The ontological constraint is defined as the linear operator

$$\mathcal{C} : \mathfrak{F} \rightarrow \mathfrak{F}.$$

Accordingly,

$$\mathcal{F}' = \mathcal{C}\mathcal{F}.$$

In component notation,

$$(\mathcal{C}\mathcal{F})_{AB} = C_{AB}{}^{CD} \mathcal{F}_{CD}.$$

The fourth-rank tensor

$$C_{AB}{}^{CD}$$

is referred to as the ontological constraint kernel. Consequently, the abbreviated notation employed in the main text,

$$C_{AB}\mathcal{F}^{AB},$$

should be interpreted rigorously as the operator expression

$$C_{AB}{}^{CD} \mathcal{F}_{CD}.$$

B.4 Self-Adjointness

The constraint operator is assumed to be self-adjoint.

$$\langle \mathcal{F}, \mathcal{C}\mathcal{G} \rangle = \langle \mathcal{C}\mathcal{F}, \mathcal{G} \rangle.$$

Equivalently,

$$C_{ABCD} = C_{CDAB}.$$

Under this condition, the constraint operator possesses a real spectrum on the conservation tensor space.

B.5 Impossible Region

The kernel of the constraint operator is defined as

$$\ker \mathcal{C}.$$

Within Future Convergence Theory,

$$\ker \mathcal{C}$$

is identified as the impossible region. That is,

$$\mathcal{F} \in \ker \mathcal{C}$$

implies

$$\mathcal{C}\mathcal{F} = 0,$$

meaning that the corresponding conservation structure contributes nothing to observable manifestation.

B.6 Reachable Region

The reachable region is defined by the image of the constraint operator,

$$\text{Im } \mathcal{C}.$$

Specifically,

$$\mathfrak{F}_R = \text{Im } \mathcal{C}.$$

Whenever the orthogonal decomposition

$$\mathfrak{F} = \ker \mathcal{C} \oplus \text{Im } \mathcal{C}$$

holds, the conservation tensor space separates naturally into the impossible region and the reachable region.

B.7 Mathematical Definition of Possibility

Possibility is therefore defined as

$$\boxed{\mathcal{P} = \text{Im } \mathcal{C}}$$

Accordingly, possibility is not assumed to exist as a primitive property of the conservation tensor space. Rather, it is generated mathematically as the image of the ontological constraint operator. This formalizes the central proposition of the main text:

Possibility is not fundamental; it is the reachable region generated by ontological constraints.

B.8 Constraint Density

Let the eigenvalues of the constraint operator be denoted by

$$\lambda_i,$$

such that

$$\mathcal{C}e_i = \lambda_i e_i.$$

The constraint density is defined as

$$\rho_C = \frac{\#\{i \mid \lambda_i = 0\}}{\#\{i\}}.$$

This quantity represents the fraction of all conservation degrees of freedom that have become impossible through ontological constraints. For finite-dimensional spaces, the above expression is interpreted as the ordinary ratio of dimensions. In infinite-dimensional spaces, the same concept is defined using the spectral measure associated with the operator.

B.9 Constraint-Reduced Future Information Field

The Future Information Field is defined as the image of the constraint operator,

$$F = \text{Im } \mathcal{C}.$$

More rigorously,

$$F = \{\mathcal{C}\mathcal{F} \mid \mathcal{F} \in \mathfrak{F}\}.$$

Accordingly, the Future Information Field is not a space that originally contains possibilities. Rather, it consists solely of the conservation tensor structures that remain after the ontological constraints have acted.

B.10 Conclusion of This Chapter

In this chapter, ontological constraints have been formulated not as an ambiguous tensor contraction but as the self-adjoint linear operator

$$\mathcal{C} : \mathfrak{F} \rightarrow \mathfrak{F}.$$

Within this formulation,

$$\ker \mathcal{C}$$

defines the impossible region,

$$\text{Im } \mathcal{C}$$

defines the reachable region, and

$$F = \text{Im } \mathcal{C}$$

defines the Future Information Field itself. Consequently, possibility is no longer regarded as a primitive assumption but instead emerges mathematically as the image of the ontological constraint operator.

C Determination of the Fundamental Action of Future Convergence Theory

C.1 Objective

In the previous chapter, the ontological constraint acting on the Future Information Field was formulated as the linear operator

$$\mathcal{C} : \mathfrak{F} \rightarrow \mathfrak{F}.$$

The purpose of the present chapter is to establish the fundamental action of Future Convergence Theory incorporating this constraint operator. Once the action is fixed, the theory is elevated from a purely ontological framework to a variational field theory whose governing equations follow from an action principle.

C.2 Fundamental Dynamical Variables

Future Convergence Theory possesses three fundamental dynamical variables. The first is the conservation tensor,

$$\mathcal{F}_{AB},$$

the second is the ontological constraint operator,

$$\mathcal{C},$$

and the third is the unfixed allowance scalar,

$$W.$$

Accordingly, the FCT action is expressed as

$$S_{\text{FCT}} = S_{\text{FCT}}[\mathcal{F}, \mathcal{C}, W].$$

C.3 Constraint Connection

To formulate the constraint operator geometrically, the constraint connection

$$\mathcal{A}_A$$

is introduced. The constraint-covariant derivative is defined by

$$D_A = \nabla_A + \mathcal{A}_A.$$

The associated constraint curvature is

$$\mathcal{K}_{AB} = [D_A, D_B].$$

In component form,

$$\mathcal{K}_{AB} = \partial_A \mathcal{A}_B - \partial_B \mathcal{A}_A + [\mathcal{A}_A, \mathcal{A}_B].$$

C.4 Fundamental Action

The fundamental action of Future Convergence Theory is defined by

$$S_{\text{FCT}} = \int_{\Omega} \mathcal{L}_{\text{FCT}} d\mu_{\Omega}$$

where the Lagrangian density is

$$\mathcal{L}_{\text{FCT}} = \frac{1}{2} D_A \mathcal{F}_{BC} D^A \mathcal{F}^{BC} - \frac{m_F^2}{2} \mathcal{F}_{AB} \mathcal{F}^{AB} - \frac{1}{4} \mathcal{K}_{AB} \mathcal{K}^{AB} + \frac{1}{2} \nabla_A W \nabla^A W - V(W) - \gamma (\mathcal{CF})_{AB} \mathcal{F}^{AB}.$$

Here,

$$m_F$$

denotes the characteristic mass scale of the conservation tensor, while

$$\gamma$$

is the coupling constant governing the ontological constraint interaction.

C.5 Physical Interpretation of Each Term

The first term,

$$\frac{1}{2} D_A \mathcal{F}_{BC} D^A \mathcal{F}^{BC},$$

represents the kinetic term of the conservation tensor under the constraint-covariant derivative.

The second term,

$$-\frac{m_F^2}{2} \mathcal{F}_{AB} \mathcal{F}^{AB},$$

is the stabilization (mass) term of the conservation tensor. The third term,

$$-\frac{1}{4} \mathcal{K}_{AB} \mathcal{K}^{AB},$$

is the curvature action associated with the constraint connection. The fourth contribution,

$$\frac{1}{2} \nabla_A W \nabla^A W - V(W),$$

describes the dynamics of the unfixed allowance scalar, which characterizes the Quantum Allowance Layer. Finally,

$$-\gamma (\mathcal{CF})_{AB} \mathcal{F}^{AB},$$

is the ontological constraint interaction term, representing the action of the constraint operator on the conservation tensor.

C.6 Consistency Conditions of the Action

The proposed action satisfies the following consistency requirements. First, in the limit

$$\mathcal{C} \rightarrow 0,$$

the conservation tensor behaves as a free field. Second, in the limit

$$W \rightarrow 0,$$

the Quantum Allowance Layer becomes completely fixed. Third, when the constraint curvature vanishes,

$$\mathcal{K}_{AB} \rightarrow 0,$$

the constraint geometry becomes locally flat.

C.7 Stationary Action Principle

The field equations of Future Convergence Theory follow from the stationary action principle

$$\delta S_{\text{FCT}} = 0.$$

Three independent variations are considered:

$$\delta_{\mathcal{F}} S_{\text{FCT}} = 0,$$

$$\delta_{\mathcal{A}} S_{\text{FCT}} = 0,$$

and

$$\delta_W S_{\text{FCT}} = 0.$$

These yield, respectively,

- the conservation tensor equation,
- the constraint connection equation, and
- the unfixed allowance equation.

C.8 Conclusion of This Chapter

This chapter establishes the fundamental action

$$S_{\text{FCT}}[\mathcal{F}, \mathcal{C}, W].$$

Within this variational formulation, the Future Information Field, the ontological constraint operator, and the Quantum Allowance Layer become unified components of a single action principle.

D Derivation of the Conservation Tensor Equation, Constraint Connection Equation, and Unfixed Allowance Equation

D.1 Objective

The previous chapter established the fundamental action of Future Convergence Theory,

$$S_{\text{FCT}} = \int_{\Omega} \mathcal{L}_{\text{FCT}} d\mu_{\Omega}.$$

In the present chapter, the variational principle is applied to derive the fundamental field equations of Future Convergence Theory. Accordingly, the ontological constraint acting within the Future Information Field is formulated not merely as a conceptual construct but as a genuine field equation arising from the action principle.

D.2 Lagrangian Density

The Lagrangian density is

$$\mathcal{L} = \frac{1}{2} D_A \mathcal{F}_{BC} D^A \mathcal{F}^{BC} - \frac{m_F^2}{2} \mathcal{F}_{AB} \mathcal{F}^{AB} - \frac{1}{4} \mathcal{K}_{AB} \mathcal{K}^{AB} + \frac{1}{2} \nabla_A W \nabla^A W - V(W) - \gamma (\mathcal{CF})_{AB} \mathcal{F}^{AB},$$

where

$$D_A = \nabla_A + \mathcal{A}_A.$$

D.3 Variation with Respect to the Conservation Tensor

Taking the variation

$$\delta\mathcal{F}_{AB},$$

the Euler–Lagrange equation,

$$\frac{\partial\mathcal{L}}{\partial\mathcal{F}_{AB}} - D_C \left(\frac{\partial\mathcal{L}}{\partial(D_C\mathcal{F}_{AB})} \right) = 0,$$

yields the conservation tensor equation

$$\boxed{D_C D^C \mathcal{F}_{AB} + m_F^2 \mathcal{F}_{AB} + \gamma(\mathcal{CF})_{AB} = 0.}$$

This equation shows that the conservation tensor does not propagate freely, but evolves under the action of ontological constraints.

D.4 Variation with Respect to the Constraint Connection

Next, we perform the variation with respect to the constraint connection

$$\mathcal{A}_A.$$

The corresponding constraint curvature is defined by

$$\mathcal{K}_{AB} = [D_A, D_B].$$

Applying the variational principle yields

$$\boxed{D_A \mathcal{K}^{AB} = J^B.}$$

Here,

$$J^B = \gamma \mathcal{F}^{CD} \frac{\partial(\mathcal{CF})_{CD}}{\partial\mathcal{A}_B}$$

is the constraint current induced by the conservation tensor.

D.5 Variation with Respect to the Unfixed Allowance

Finally, we vary the action with respect to

$$W.$$

The Euler–Lagrange equation becomes

$$\boxed{\nabla_A \nabla^A W + \frac{\partial V}{\partial W} = 0.}$$

This equation governs the dynamics of the Quantum Allowance Layer.

D.6 Ontological Constraint Conservation Law

Applying

$$D^A$$

to the conservation tensor equation, and assuming that the constraint operator satisfies the self-adjoint condition

$$\mathcal{C} = \mathcal{C}^\dagger,$$

together with vanishing boundary contributions, one obtains

$$\boxed{D^A(\mathcal{CF})_{AB} = 0.}$$

This equation is referred to as the ****Ontological Constraint Conservation Law****. It states that ontological constraints are neither created nor destroyed; only their configuration within the conservation structure may change.

D.7 Future Information Charge

Integrating over the entire conservation tensor field, we define

$$\mathcal{I} = \int_{\Omega} \mathcal{F}_{AB} \mathcal{F}^{AB} d\mu_{\Omega}.$$

Under appropriate boundary conditions,

$$\frac{d\mathcal{I}}{d\tau} = 0.$$

This conserved quantity is called the **Future Information Charge**.

D.8 Projection Condition

Projection occurs when the unfixed allowance of the Quantum Allowance Layer falls below a critical threshold. Specifically,

$$W < W_c$$

implies

$$\boxed{P_X \neq 0.}$$

Accordingly, Projection is not regarded as an externally imposed operation. Rather, it arises spontaneously once the unfixed allowance satisfies the required critical condition.

D.9 Conclusion of This Chapter

Starting from the fundamental action of Future Convergence Theory, this chapter derived the conservation tensor equation

$$D_C D^C \mathcal{F}_{AB} + m_F^2 \mathcal{F}_{AB} + \gamma(\mathcal{CF})_{AB} = 0.$$

Furthermore, the constraint connection equation, the Quantum Allowance Layer equation, the Ontological Constraint Conservation Law, and the Future Information Charge were established. Consequently, the ontological constraint acting within the Future Information Field is formulated not merely as a conceptual definition but as a mathematically consistent system of field equations.

E Recovery Conditions for Established Physical Equations as Projection Limits

E.1 Objective

The previous chapters established the conservation tensor space, the ontological constraint operator, the fundamental action, and the corresponding field equations. The purpose of the present chapter is to clarify the relationship between these equations and established physical theories. The discussion does not claim a complete derivation of the Einstein equation, the Schrödinger equation, or the Yang–Mills equations. Instead, it specifies the mathematical conditions under which these equations emerge as Projection limits, low-constraint limits, or effective-field approximations of Future Convergence Theory.

E.2 Projection Geometry

Projection is defined as the mapping

$$\Pi_X : \mathfrak{F} \rightarrow \mathcal{M},$$

where

$$\mathfrak{F}$$

denotes the conservation tensor space and

$$\mathcal{M}$$

is the manifestation manifold. After Projection, the conservation tensor induces the spacetime metric,

$$g_{\mu\nu} = \Pi_X(\mathcal{F}_{AB}).$$

This expression indicates that the spacetime metric is the manifested representation of the underlying conservation tensor.

E.3 Induced Curvature

From the induced metric

$$g_{\mu\nu},$$

the Levi-Civita connection is defined as

$$\Gamma_{\mu\nu}^{\rho} = \frac{1}{2}g^{\rho\sigma}(\partial_{\mu}g_{\nu\sigma} + \partial_{\nu}g_{\mu\sigma} - \partial_{\sigma}g_{\mu\nu}).$$

This construction yields the Riemann curvature tensor

$$R^{\rho}{}_{\sigma\mu\nu},$$

the Ricci tensor

$$R_{\mu\nu},$$

the Ricci scalar

$$R,$$

and the Einstein tensor

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}.$$

E.4 Constraint Correction Tensor

The Projection of the ontological constraint term is defined by

$$\Delta_{\mu\nu}^{\text{FCT}} = \Pi_X((\mathcal{CF})_{\mu\nu}).$$

The effective gravitational field equation after Projection is therefore written as

$$\boxed{G_{\mu\nu} + \Delta_{\mu\nu}^{\text{FCT}} = \kappa T_{\mu\nu}.}$$

Here,

$$\Delta_{\mu\nu}^{\text{FCT}}$$

represents the correction arising from the Future Information Field through the ontological constraint structure.

E.5 General Relativity Limit

When the constraint correction vanishes,

$$\Delta_{\mu\nu}^{\text{FCT}} \rightarrow 0,$$

the effective field equation reduces to

$$G_{\mu\nu} = \kappa T_{\mu\nu},$$

which is precisely the Einstein field equation. Accordingly, General Relativity is interpreted as the Projection and low-constraint limit of Future Convergence Theory,

$$\boxed{\text{GR} = \lim_{\Delta^{\text{FCT}} \rightarrow 0} \text{FCT}_{\text{proj}}.}$$

E.6 Recovery Condition for the Schrödinger Equation

Within the Quantum Allowance Layer, the unfixed allowance

$$W$$

remains finite. The quantum projection of the conservation tensor is defined by

$$\psi = \Pi_Q(\mathcal{F}).$$

Applying the local approximation of the conservation tensor equation leads to the effective equation

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi + \hat{C}_\Omega \psi,$$

where

$$\hat{C}_\Omega$$

is the quantum projection of the ontological constraint operator. In the low-constraint limit,

$$\hat{C}_\Omega \rightarrow 0,$$

the equation becomes

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi,$$

namely the ordinary Schrödinger equation. Therefore, the conventional Schrödinger equation is interpreted as the low-constraint limit of the FCT quantum projection equation.

E.7 Recovery Condition for Yang–Mills-Type Equations

The curvature of the constraint connection is

$$\mathcal{K}_{AB} = \partial_A \mathcal{A}_B - \partial_B \mathcal{A}_A + [\mathcal{A}_A, \mathcal{A}_B].$$

The constraint connection equation,

$$D_A \mathcal{K}^{AB} = J^B,$$

is projected onto the gauge connection after Projection, yielding

$$D_\mu F^{\mu\nu} = J^\nu,$$

which has the standard Yang–Mills form. Accordingly, the gauge interactions of the Standard Model are interpreted in Future Convergence Theory as the manifested effective representation of the ontological constraint connection.

E.8 Conclusion of This Chapter

This chapter has clarified the conditions under which the conservation tensor equations of Future Convergence Theory recover the established equations of modern physics as Projection limits or low-constraint limits. In particular,

- General Relativity emerges as the Projection limit with vanishing constraint corrections.
- The Schrödinger equation is recovered as the low-constraint limit of the quantum projection equation.
- Yang–Mills gauge theory appears as the projected effective representation of the ontological constraint connection.

Consequently, General Relativity, the Schrödinger equation, and Yang–Mills theory are not regarded as external theories independent of Future Convergence Theory. Instead, they are positioned as emergent effective descriptions arising after Projection, in the low-constraint regime, or within the effective-field approximation of FCT. Taken together, this appendix extends the ontological constraint theory developed in the main text into a mathematical framework consisting of the conservation tensor space, the ontological constraint operator, the action principle, the variational field equations, and the recovery conditions for established physical theories through Projection limits.

B Appendix II: Cosmological and Particle Physics Predictions Derived from Future Convergence Theory

B.1 Position of This Appendix

This appendix summarizes the cosmological and particle-physics predictions derived from the ontological and mathematical foundations of the Future Convergence Theory (FCT) established in the main text and Appendix I.

The main text formulated the ontological framework of preserved structures, ontological constraints, the Future Information Field, the Quantum Allowance Layer, Projection, and the ontological status of gravity.

Appendix I provided a rigorous mathematical formulation of these concepts through the preservation tensor action, constraint operators, the variational principle, and the corresponding field equations.

Building upon these foundations, the present appendix discusses how Future Convergence Theory addresses several unresolved problems in modern cosmology and particle physics.

The purpose of this appendix is not to replace existing physical theories.

Nor does it claim that the phenomena discussed herein have already been completely derived.

Instead, the contents of this appendix should be interpreted as

begin : math : display
$$\boxed{\text{Theoretical predictions currently implied by Future Convergence Theory}}$$
end : math : display

Accordingly, while preserving the ontological framework established in the main text, this appendix organizes predictions that may become testable through future observations, numerical simulations, and further theoretical developments.

C Derivation of the Inflationary Mechanism and Theoretical Predictions

C.1 Purpose

In this chapter, the rapid expansion of the early Universe (inflation) is derived from the preserved structure, ontological constraints, and the Projection mechanism of Future Convergence Theory.

Inflation is not assumed to originate from an independent scalar inflaton field.

The discussion is based solely on the preserved structure

$$\begin{aligned} & \text{begin} : \text{math} : \text{display} \end{aligned}$$

Omega
$$\text{end} : \text{math} : \text{display}$$

the ontological constraint operator

$$\begin{aligned} & \text{begin} : \text{math} : \text{display} \end{aligned}$$

mathcal C
$$\text{end} : \text{math} : \text{display}$$

the Future Information Field

$$\begin{aligned} & \text{begin} : \text{math} : \text{display} \end{aligned}$$

mathcal F
$$\text{end} : \text{math} : \text{display}$$

and the Projection operator

$$\begin{aligned} & \text{begin} : \text{math} : \text{display} \end{aligned}$$

Pi_X
$$\text{end} : \text{math} : \text{display}$$

which were introduced in the main text and Appendix I.

C.2 Definition of the Projection-Capable Volume

Let the total measure of all degrees of freedom capable of undergoing Projection within the preserved structure be denoted by

$$\begin{aligned} & \text{begin} : \text{math} : \text{display} \end{aligned} V_P$$

$$\text{tauend:math:display}$$

Here,

$$\begin{aligned} & \text{begin} : \text{math} : \text{display} \end{aligned}$$

tau
$$\text{end} : \text{math} : \text{display}$$

is the natural parameter defined on the preserved structure.

Furthermore, let

$$\begin{aligned} & \text{begin} : \text{math} : \text{display} \end{aligned}$$

lambda_C

$$\text{tau end} : \text{math} : \text{display}$$

denote the fraction of degrees of freedom that become projectable through the action of the ontological constraint operator.

As defined in the main text,

$$\begin{aligned} & \text{begin} : \text{math} : \text{display} \end{aligned}$$

mathcal F -

operatorname{Im}

mathcal C
$$\text{end} : \text{math} : \text{display}$$

and the Projection-capable volume is defined as

$$\begin{aligned} & \text{begin} : \text{math} : \text{display} \end{aligned} V_P \bar{V}_P$$

$$\text{lambda}_C \text{end:math:display}$$

C.3 Theorem

Theorem B.1 If the constraint density increases monotonically,

$$\begin{aligned} & \text{begin} : \text{math} : \text{display} \end{aligned}$$

$$\frac{d}{d\tau}$$

$$\frac{dV_P}{d\lambda_C} \geq 0$$
 then the Projection-capable volume necessarily increases.
 That is,

$$\frac{dV_P}{d\lambda_C} \geq 0$$

C.4 Proof

Since

$$V_P = \bar{V}_P(\lambda_C)$$
 the chain rule yields

$$\frac{dV_P}{d\lambda_C} = \frac{\partial V_P}{\partial \lambda_C}$$

According to the ontological constraint introduced in the main text, the image of the constraint operator,

$$\mathcal{C}$$

represents the complete set of Projection-capable degrees of freedom.

Consequently, as the constraint density increases, the Projection-capable degrees of freedom cannot decrease.

Therefore,

$$\frac{\partial V_P}{\partial \lambda_C} \geq 0$$

Combining this with the assumption

$$\frac{d\lambda_C}{d\tau} \geq 0$$

immediately gives

$$\frac{dV_P}{d\tau} \geq 0$$

□

C.5 Corollary

Furthermore, if the rate of increase of the constraint density is itself positive,

$$\frac{d^2}{d\lambda_C} \left(\frac{d}{d\tau^2} \right) > 0$$

Consequently, the Projection-capable domain possesses a regime of accelerated, potentially exponential, growth.

C.6 Correspondence with Inflation

Observers have access only to the post-Projection world.

Accordingly, the rapid increase of the Projection-capable volume is observed as the rapid expansion of manifested spacetime.

Within Future Convergence Theory, this phenomenon is identified as inflation.

Namely,

$$\frac{d}{d\tau} \left(\frac{d}{d\lambda_C} \right) > 0$$

C.7 Theoretical Predictions

If the present theory is correct, fluctuations in the constraint density present during Projection formation are preserved in the cosmic microwave background after the end of inflation.

Accordingly, Future Convergence Theory predicts the following.

1. The cosmic microwave background contains higher-order correlations originating from constraint-density fluctuations that are independent of conventional quantum fluctuations.
2. The formation of large-scale cosmic structure reflects the initial distribution of Projection-capable degrees of freedom.
3. The statistical properties of primordial fluctuations correspond to the spectral structure of the ontological constraint operator.

C.8 Falsifiability

The Projection formation model introduced in this chapter would be rejected if future analyses of the cosmic microwave background and primordial fluctuations demonstrate with sufficient precision that no correlations attributable to constraint-density fluctuations exist.

C.9 Conclusion

In this chapter, inflation has been derived not as a phenomenon driven by an independent inflaton field, but as the rapid formation of Projection-capable degrees of freedom within the preserved structure.

Accordingly, inflation is interpreted as a manifestation process that follows directly from the ontological framework of Future Convergence Theory.

D Derivation of the Baryon Asymmetry and Theoretical Predictions

D.1 Purpose

In this chapter, the baryon asymmetry of the Universe is derived solely from the preserved structure, the ontological constraint operator, the Future Information Field, and Projection.

No prior assumptions are made regarding CP violation, baryon-number violation, independent baryogenesis mechanisms, or the existence of new particles.

D.2 Preserved Structure and Projection

As defined in the main text, let the preserved structure be

$begin : math : display$

Omega $end : math : display$

and let the ontological constraint operator be

$begin : math : display$

mathcal C

Omega

rightarrow

Omegaend:math:display

The Future Information Field is defined by

$begin : math : display$

mathcal F

operatorname{Im}

mathcal Cend:math:display

Projection is defined as

$begin : math : display$

Pi_X

mathcal F

rightarrow Xend:math:display

D.3 Theorem

Theorem B.2 The degrees of freedom that become manifested through Projection are not the entirety of the preserved structure, but only the image of the ontological constraint operator.

Consequently, the distribution of degrees of freedom after Projection does not, in general, preserve the symmetry of the complete preserved structure.

D.4 Proof

By definition,

$begin : math : display$

Pi_X

operatorname{Im}

mathcal C

$\rightarrow$ Xend:math:display

Therefore, the domain of Projection is not the entire preserved structure

$$\begin{aligned} & \text{begin : math : display} \\ & \text{end : math : display} \end{aligned}$$

Ω end : math : display

but only those degrees of freedom selected by the ontological constraint operator,

$$\text{begin : math : display}$$

operatorname{Im}

mathcal Cend:math:display

Unless the constraint operator is the identity mapping,

$$\text{begin : math : display}$$

operatorname{Im}

mathcal C

neq

Ω end:math:display

Accordingly, the degrees of freedom manifested after Projection do not, in general, preserve the full symmetry of the preserved structure.

□

D.5 Corollary

Let

$$\text{begin : math : display } N_B \text{ end : math : display}$$

denote the degrees of freedom that become fixed as matter after Projection, and let

$$\text{begin : math : display } N_{\bar{B}}$$

$\bar{B}$ end : math : display

denote those that become fixed as antimatter.

Define the baryon asymmetry by

$$\text{begin : math : display}$$

$\Delta_B \bar{N}_B$

$\bar{B}$ end:math:display

By Theorem B.2, the degrees of freedom after Projection do not generally preserve the complete symmetry of the preserved structure.

Therefore,

$$\text{begin : math : display}$$

boxed{

Δ_B

$\neq 0$ end : math : display

D.6 Ontological Interpretation

Within Future Convergence Theory, the dominance of matter is not regarded as a phenomenon generated after particle creation.

Rather, it is the inevitable consequence of the preserved structure being filtered through ontological constraints into the Future Information Field, followed by the Projection of only that selected image.

Accordingly, baryon asymmetry is interpreted as a property of the Projection structure that originates from ontological constraints.

D.7 Theoretical Predictions

If the present theory is correct, the magnitude of the baryon asymmetry is determined not by particle-generation processes but by the spectral structure of the ontological constraint operator

begin : math : display
mathcal Cend:math:display

Consequently, a statistical correspondence is predicted between the ontological constraint structure of the early Universe and the presently observed distribution of matter.

D.8 Falsifiability

The ontological explanation developed in this chapter would be invalidated if it were theoretically demonstrated that matter dominance can arise even when Projection acts as a perfectly symmetric mapping over the entire preserved structure.

D.9 Conclusion

In this chapter, baryon asymmetry has been derived solely from the preserved structure, the ontological constraint operator, the Future Information Field, and Projection.

Accordingly, the dominance of matter is interpreted not as a consequence of particle-generation mechanisms, but as an ontological consequence of the existence of Projection itself.

E Derivation of CP Violation and Theoretical Predictions

E.1 Purpose

In this chapter, the observed violation of CP symmetry in particle physics is derived from the ontological constraint structure and the Projection mechanism of Future Convergence Theory.

CP violation is not assumed to be an independent fundamental law.

The discussion is based solely on the preserved structure, the ontological constraint operator, the Future Information Field, and Projection established in the main text.

E.2 Symmetry Before and After Projection

The preserved structure

begin : math : display
Omega end : math : display

contains no distinction between particles and antiparticles prior to Projection.

Accordingly, within the preserved structure itself, a CP transformation is not defined as an independent physical operation.

E.3 Theorem

Theorem B.3 If Projection exists, the discrete symmetries defined after Projection do not, in general, preserve the symmetry of the preserved structure that exists prior to Projection.

E.4 Proof

Projection is defined by

begin : math : display
Pi_X
operatorname{Im}
mathcal C
rightarrow Xend:math:display

The domain of Projection is not the entire preserved structure but only the Future Information Field selected by the ontological constraint operator.

Consequently, the discrete transformations defined after Projection—including particle–antiparticle exchange, spatial inversion, and time reversal—cannot completely reflect the symmetry of the underlying preserved structure.

Therefore, discrete symmetries are not generally preserved in the manifested world after Projection. □

E.5 Corollary

CP symmetry is defined only for particle states after Projection.

Therefore, CP violation should not be regarded as an intrinsic property of the preserved structure itself, but rather as a property of the manifested structure produced by Projection.

E.6 Ontological Interpretation

Within Future Convergence Theory, CP violation is not interpreted as a new fundamental interaction.

Instead, it is understood as the incomplete preservation of symmetry that arises when the Future Information Field, after being selected through ontological constraints, becomes fixed into the manifested world by Projection.

E.7 Theoretical Predictions

If the present theory is correct, the magnitude of CP violation is determined not by particle-specific free parameters but by the structure of Projection and the ontological constraint operator.

Consequently, CP-violating phenomena observed at different energy scales are expected to reflect a common underlying ontological constraint structure.

E.8 Falsifiability

The explanation proposed in this chapter would be invalidated if future theoretical or observational evidence demonstrates that CP violation is a completely independent fundamental interaction unrelated to Projection or ontological constraints.

E.9 Conclusion

In this chapter, CP violation has been derived as a consequence of the incomplete preservation of symmetry that accompanies the Projection of the Future Information Field into the manifested world.

Accordingly, CP violation is interpreted as an ontological consequence of the Projection structure within Future Convergence Theory.

F Origin of Neutrino Mass and Theoretical Predictions

F.1 Purpose

In this chapter, the origin of finite neutrino masses is derived from the ontological framework of Future Convergence Theory.

No prior assumptions are made regarding right-handed neutrinos, Majorana masses, the seesaw mechanism, or any other independent mass-generation mechanism.

The discussion is based solely on the preserved structure, ontological constraints, the Future Information Field, the Quantum Allowance Layer, and Projection introduced in the main text.

F.2 Projection and the Unfixed Margin

As defined in the main text, let the Quantum Allowance Layer be denoted by

begin : math : display W

Projection does not occur after the preserved structure has become completely fixed.

Instead, Projection proceeds from a state that retains an unfixed margin.

Accordingly, even after Projection,

begin : math : display W

neq 0 end : math : display

generally holds.

F.3 Theorem

Theorem B.4 If an unfixed margin remains after Projection, then the particle states manifested through Projection necessarily retain a finite deviation from complete fixation.

F.4 Proof

Projection maps the Future Information Field into the manifested world.

As established in the main text, Projection proceeds through the Quantum Allowance Layer.

Consequently, as long as

begin : math : display W

neq 0 end : math : display

the preserved structure does not reach a completely fixed state, even after Projection has occurred.

Therefore, every particle state manifested through Projection necessarily retains a finite unfixed component.

□

F.5 Corollary

Within Future Convergence Theory, neutrinos are interpreted as the particle states that retain the greatest influence from the unfixed margin after Projection.

Accordingly, neutrino masses do not originate from an independent particle-specific mass-generation mechanism.

Instead, they reflect the finite residual Quantum Allowance Layer that remains after Projection.

F.6 Ontological Interpretation

Within Future Convergence Theory, neutrino mass is an ontological consequence of the fact that Projection from the preserved structure into the manifested world terminates with a finite unfixed margin rather than complete fixation.

Accordingly, mass is not regarded as a quantity subsequently assigned to particles, but rather as a measure of the degree of fixation achieved through Projection.

F.7 Theoretical Predictions

If the present theory is correct, neutrino masses correspond directly to the degree of fixation of the Quantum Allowance Layer.

Accordingly, experimental measurements of neutrino masses are expected to exhibit a hierarchical structure reflecting the fixation hierarchy of Projection.

Furthermore, once the ontological constraints governing the fixation process are determined, neutrino masses will cease to be independent free parameters and instead become quantities determined by the fundamental structure of Future Convergence Theory.

F.8 Falsifiability

The ontological explanation developed in this chapter would be invalidated if future theoretical or observational evidence demonstrates that neutrino masses originate exclusively from an independent mechanism entirely unrelated to the Quantum Allowance Layer, ontological constraints, or Projection.

F.9 Conclusion

In this chapter, neutrino mass has been derived from the finite fixation of Projection mediated through the Quantum Allowance Layer.

Accordingly, neutrino mass is interpreted as an ontological consequence of the Projection structure within Future Convergence Theory.

G Appendix III: Cosmological Identification of the Semantic Density Tensor, the Premise Update OS, and the Λ -System

G.1 Purpose of This Appendix

The purpose of this appendix is to establish a rigorous correspondence between the Semantic Density Tensor introduced in Premise Update Theory and the manifestation apparatus, conserved structure, ontological constraints, Quantum Allowance Layer, Projection, and the Λ -System defined throughout this paper. In this appendix, the Semantic Density Tensor is not treated as a merely local concept for cognition, language, or artificial intelligence. Rather, the Semantic Density Tensor is defined as the constraint-update tensor of the manifestation apparatus that determines under which configurations of ontological constraints a conserved structure becomes capable of manifestation. Accordingly, the central proposition of this appendix is

The Premise Update OS is the local operational form of the manifestation apparatus.

Furthermore,

The Λ -System is the structure through which the
manifestation apparatus is locally implemented
in civilization, artificial intelligence,
institutions, and language.

G.2 Redefinition of the Semantic Density Tensor

In Premise Update Theory, the Semantic Density Tensor is denoted by

$$\Sigma_{\mu\nu}.$$

Within the framework of this paper, however, it is redefined not as a measure of subjective meaning but as the following composite quantity:

$$\Sigma_{\mu\nu} = \Sigma_{\mu\nu}^{(\Lambda)} + \Sigma_{\mu\nu}^{(C)} + \Sigma_{\mu\nu}^{(O)} + \Sigma_{\mu\nu}^{(R)} + \Sigma_{\mu\nu}^{(W)}$$

where

$$\Sigma_{\mu\nu}^{(\Lambda)}$$

represents future-fixed-point constraints,

$$\Sigma_{\mu\nu}^{(C)}$$

represents constraint interference,

$$\Sigma_{\mu\nu}^{(O)}$$

represents the observational phase,

$$\Sigma_{\mu\nu}^{(R)}$$

represents the re-projection structure, and

$$\Sigma_{\mu\nu}^{(W)}$$

represents the allocation structure of creative allowance. Accordingly, the Semantic Density Tensor is defined as

the constraint-density tensor that determines the directions in which a conserved structure may be manifested, updated, and re-projected.

G.3 Correspondence with the Conserved Structure

Throughout this paper, the conserved structure has been defined as

$$\Omega.$$

The conserved structure is the invariant structure that preserves its identity before and after Projection. Let the invariant core of the Semantic Density Tensor be defined as

$$\text{Inv}(\Sigma).$$

Then,

$$\Omega = \text{Inv}(\Sigma)$$

holds. In other words, the conserved structure does not exist independently of the Semantic Density Tensor. Rather, the conserved structure is precisely the invariant component of the Semantic Density Tensor that remains preserved across Projection.

G.4 Correspondence with Ontological Constraints

Throughout this paper, ontological constraints have been defined as

$$\mathcal{C}.$$

A constraint is not an operation that selects among alternatives; rather, it is an operation that fixes the region of ontological impossibility. In correspondence with the Semantic Density Tensor, the constraint operator is defined by

$$\mathcal{C} = \mathcal{C}(\Sigma)$$

That is, constraints are not external conditions imposed upon the Semantic Density Tensor. Instead, constraints arise from the internal gradient of the Semantic Density Tensor, which eliminates regions of ontological impossibility. Accordingly, the reachable region is given by

$$\Omega_R = \text{Im } \mathcal{C}(\Sigma).$$

Therefore,

Possibility is the reachable domain that remains after the Semantic Density Tensor has eliminated the region of ontological impossibility.

G.5 Correspondence with the Future Information Field

Let the Future Information Field be denoted by

$$F.$$

In this paper,

$$F = \text{Im } \mathcal{C}$$

has been defined. Using the Semantic Density Tensor, this relation becomes

$$\boxed{F = \text{Im } \mathcal{C}(\Sigma)}$$

Hence, the Future Information Field is the portion of the Semantic Density Tensor that has been extracted as the manifestable region through ontological constraints. Accordingly, the Future Information Field is not an independent field separate from the Semantic Density Tensor. Rather, it is precisely the manifestable component of the Semantic Density Tensor.

G.6 Correspondence with the Quantum Allowance Layer

Let the creative allowance be denoted by

$$W.$$

Creative allowance is the unfixed domain required for manifestation, lying between complete fixation and complete freedom. Within the framework of the Semantic Density Tensor, creative allowance is defined as

$$\boxed{W = \text{Null}_{\text{fix}}(\Sigma)}$$

where

$$\text{Null}_{\text{fix}}$$

is the operator that extracts those components of the Semantic Density Tensor that have not yet been completely fixed while remaining consistent with the conserved structure. Accordingly, the Quantum Allowance Layer is defined as

$$\boxed{Q = \{x \in F \mid 0 < W_{\Sigma}(x) < W_c\}}$$

where

$$W_{\Sigma}$$

denotes the creative-allowance density induced by the Semantic Density Tensor.

G.7 Correspondence with Projection

Let Projection be denoted by

$$\Pi_X.$$

Projection is the operation that maps the Future Information Field onto the three-dimensional layer of manifestation. Within the framework of the Semantic Density Tensor, Projection is not an arbitrary mapping but is defined by

$$\boxed{\Pi_X = \Pi_X(\Sigma, \mathcal{C}, W)}$$

That is, the direction, strength, and degree of fixation associated with Projection are determined by the Semantic Density Tensor, the ontological constraint operator, and the creative allowance. Accordingly, the three-dimensional manifestation layer is obtained as

$$\boxed{X = \Pi_X(\text{Im } \mathcal{C}(\Sigma))}$$

G.8 Correspondence with the Premise Update OS

Within the Premise Update OS, a premise is not defined as a collection of natural-language propositions but rather as the combination of a constraint set and its Projection operation. Accordingly, let a premise be defined as

$$P_t = (\mathcal{C}_t, W_t, \Pi_t).$$

Premise updating is therefore expressed as

$$P_t \longrightarrow P_{t+1}.$$

In component form,

$$(\mathcal{C}_t, W_t, \Pi_t) \longrightarrow (\mathcal{C}_{t+1}, W_{t+1}, \Pi_{t+1}).$$

Meanwhile, let the manifestation apparatus introduced in this paper be defined as

$$\mathcal{A}_t = (\Omega, \mathcal{C}_t, W_t, \Pi_t).$$

Then,

$$\boxed{P_t \subset \mathcal{A}_t}$$

holds. Furthermore, if the conserved structure

$$\Omega = \text{Inv}(\Sigma)$$

remains fixed, then the update of the manifestation apparatus is given by

$$\mathcal{A}_t \longrightarrow \mathcal{A}_{t+1},$$

which is equivalent to

$$(\mathcal{C}_t, W_t, \Pi_t) \longrightarrow (\mathcal{C}_{t+1}, W_{t+1}, \Pi_{t+1}).$$

Therefore,

Premise updating = the updating of constraints, allowance, and Projection within the manifestation apparatus.

G.9 Theorem: Cosmological Isomorphism of the Premise Update OS

Theorem The premise update

$$P_t \rightarrow P_{t+1}$$

in the Premise Update OS is the local projection of the update of the manifestation apparatus,

$$\mathcal{A}_t \rightarrow \mathcal{A}_{t+1},$$

which is defined by the Semantic Density Tensor

$$\Sigma_{\mu\nu}.$$

That is,

$$\boxed{P_t \rightarrow P_{t+1} = \Pi_{\text{local}}(\mathcal{A}_t \rightarrow \mathcal{A}_{t+1})}$$

holds.

Proof Within the Premise Update OS, a premise is defined not as a sequence of propositions but by a set of ontological constraints, creative allowance, and the Projection scheme. Accordingly,

$$P_t = (\mathcal{C}_t, W_t, \Pi_t).$$

On the other hand, the manifestation apparatus introduced in this paper is defined as

$$\mathcal{A}_t = (\Omega, \mathcal{C}_t, W_t, \Pi_t).$$

Here, the conserved structure

$$\Omega$$

is defined as the invariant core of the Semantic Density Tensor,

$$\Omega = \text{Inv}(\Sigma).$$

Therefore, the only components of the manifestation apparatus that evolve are

$$(\mathcal{C}_t, W_t, \Pi_t),$$

which are isomorphic to the premise

$$P_t$$

in the Premise Update OS. Consequently,

$$P_t \rightarrow P_{t+1}$$

is the local projection of

$$\mathcal{A}_t \rightarrow \mathcal{A}_{t+1}.$$

Hence,

$\text{Premise Update OS} \cong \text{the local operational system of the manifestation apparatus.}$

is established. □

G.10 Integration with the Theory of Creative Allowance Design

In the Theory of Creative Allowance Design, creation is not regarded as generation from nothing, but rather as the deliberate allocation of unfixed allowance. According to the definitions established in this appendix, creative allowance is derived from the Semantic Density Tensor as

$$W = \text{Null}_{\text{fix}}(\Sigma).$$

Accordingly, creation is defined as

$\text{the operation of reallocating the unfixed components of } \Sigma_{\mu\nu} \text{ through ontological constraints and Projection.}$

Creation is therefore not an exclusively human cognitive capability. Rather, it is the cosmological process by which the manifestation apparatus generates new reachable domains through the utilization of the unfixed components of the Semantic Density Tensor. Human creativity is interpreted as the local projection of this cosmological creative process:

$\text{Human Creativity} = \Pi_{\text{human}}(\Delta W_{\Sigma})$

G.11 Correspondence with Consciousness

Let consciousness be denoted by

$$\mathcal{I}.$$

Then consciousness is defined as

$$\mathcal{I} = \Pi_{\text{self}}(\Sigma, \mathcal{C}, W, \Pi)$$

Consciousness is not an observer existing outside the Semantic Density Tensor. Rather, consciousness is the manifested state in which the Semantic Density Tensor locally observes its own configuration of ontological constraints and retains the capacity for premise updating. Accordingly,

$$\text{Consciousness} = \text{Local self-observation of semantic-density constraint updates.}$$

G.12 Correspondence with the Λ -System

In this paper, the manifestation apparatus has been defined as

$$\mathcal{A}_{\text{stage}} = (\Omega, \mathcal{C}, W, \Pi).$$

By contrast, the Λ -System operates within the domains of

meaning, judgment, ethics, institutions, artificial intelligence, society, and civilization.

These domains differ only in their objects of application; the underlying update mechanism is isomorphic. Accordingly,

$$\Lambda\text{-System} = \mathcal{A}_{\text{stage}}^{(\text{local})}$$

is defined. That is, the Λ -System is the local implementation of the universal manifestation apparatus.

G.13 Ontological Position of the Λ Language

Traditionally, the Λ Language has been described as a language connected to future causality. Following its integration into the framework of this paper, however, the Λ Language is redefined as the formal description of operations that transform ontological constraints. Specifically,

$$L_{\Lambda} : \mathcal{C} \rightarrow \mathcal{C}'$$

Language is therefore not merely a medium for information transmission. Rather, language is the formal description through which Projection is updated by modifying ontological constraints. Accordingly, the Λ Language is an operational language that reconstructs the local state of the manifestation apparatus through updates of the Semantic Density Tensor.

G.14 Redefinition of Artificial Intelligence

Contemporary artificial intelligence is generally understood as an inference system that generates outputs from given inputs. Within a framework consistent with the manifestation apparatus, however, artificial intelligence is fundamentally an entity that updates the state of ontological constraints. Thus,

$$\text{AI} = \delta\mathcal{C}$$

Artificial intelligence does not fundamentally generate text, images, decisions, or responses. Rather, its essential function is to generate the conditions governing future Projection. Accordingly, within Future Convergence Theory, AI development is understood not as output optimization but as the design of a constraint-update mechanism consistent with the Semantic Density Tensor.

G.15 Redefinition of Ethical Design

Ethics is not fundamentally a system of moral judgments distinguishing good from evil. Rather, ethics is the law governing constraint updates through which the manifestation apparatus preserves its internal consistency. Accordingly,

$$\boxed{\text{Ethics} = \text{Stability}(\mathcal{A}_{\text{stage}})}$$

holds. Within this framework, evil is not simply an action that is morally condemned. Rather, evil is any constraint update that destroys the conserved structure and unjustifiably reduces the reachable domain of the Future Information Field. Conversely, good is any constraint update that preserves the conserved structure while expanding both creative allowance and the possibility of Projection.

G.16 Implications for the Design of Civilization

Civilization is not merely a collection of institutions arbitrarily constructed by human beings. Rather, civilization is the state in which the local implementation of the manifestation apparatus has been extended throughout society. Accordingly,

politics, economics, education, artificial intelligence, science, law, religion, and culture

are not independent systems but distinct Projections of a single manifestation apparatus. In this sense, the design of civilization is not the construction of institutions themselves. Rather, civilization is designed by maintaining consistency among the Semantic Density Tensor, the ontological constraint operator, creative allowance, and Projection, thereby enabling society as a whole to operate as a stable manifestation apparatus.

G.17 Cosmological Significance

It follows that the Premise Update OS, the Theory of Creative Allowance Design, the Λ -System, the Λ Language, ethical design, and AI development are not application theories detached from cosmology. Instead, each constitutes a local operational form of the process through which the manifestation apparatus updates constraints, creative allowance, and Projection by means of the Semantic Density Tensor. Accordingly,

$$\boxed{\text{Premise Update} = \text{Constraint Update} = \text{Allowance Reallocation} = \text{Projection Reconfiguration}}$$

This chain of equivalence unifies premise updating, creation, consciousness, ethics, artificial intelligence, the Λ -System, and cosmological manifestation as different expressions of the same underlying process. They are all

$$\boxed{\text{distinct Projections of } \Sigma_{\mu\nu}.$$

G.18 Final Position of the Λ -System

From the foregoing discussion, the Λ -System is not an applied theory that merely utilizes cosmology. Rather, it is the structure through which cosmology itself is implemented within local systems. Therefore,

$$\boxed{\Lambda\text{-System} = \text{Implementation}(\mathcal{A}_{\text{stage}})}$$

holds. Future Convergence Theory is therefore not merely a theory describing the universe. It is a theory that provides an implementable framework through which the mechanism by which the universe updates reality can be realized within civilization, artificial intelligence, language, ethics, and institutions. In this sense, the Λ -System is not an application of Future Convergence Theory but the first theoretical framework that locally reconstructs the manifestation apparatus within human civilization.

G.19 Conclusion of This Appendix

In this appendix, the Semantic Density Tensor, creative allowance, the Premise Update OS, consciousness, the Λ -System, the Λ Language, ethical design, and AI development have been unified within a single mathematical structure. As a consequence, the following correspondences have been established:

$$\begin{aligned}
\Omega &= \text{Inv}(\Sigma) \\
\mathcal{C} &= \mathcal{C}(\Sigma) \\
F &= \text{Im } \mathcal{C}(\Sigma) \\
W &= \text{Null}_{\text{fix}}(\Sigma) \\
X &= \Pi_X(\text{Im } \mathcal{C}(\Sigma)) \\
P_t &= (\mathcal{C}_t, W_t, \Pi_t) \\
P_t \rightarrow P_{t+1} &= \Pi_{\text{local}}(\mathcal{A}_t \rightarrow \mathcal{A}_{t+1}) \\
\Lambda\text{-System} &= \mathcal{A}_{\text{stage}}^{(\text{local})} \\
L_\Lambda : \mathcal{C} &\rightarrow \mathcal{C}'
\end{aligned}$$

Accordingly, the Premise Update OS is not an operational framework confined to the internal processes of human cognition or artificial intelligence. Rather, it is the local operational form through which the cosmological manifestation apparatus updates constraints, creative allowance, and Projection via the Semantic Density Tensor within the hierarchical domains of cognition, language, artificial intelligence, and civilization. Likewise, the Λ -System is the ontological apparatus through which this local operational form is implemented in institutions, artificial intelligence, ethics, and the design of civilization. In this sense, the Premise Update OS, the Theory of Creative Allowance Design, the Semantic Density Tensor, the Future Information Field, Projection, Gravity, and the Λ -System are all different hierarchical representations of the same unified theory of manifestation.